

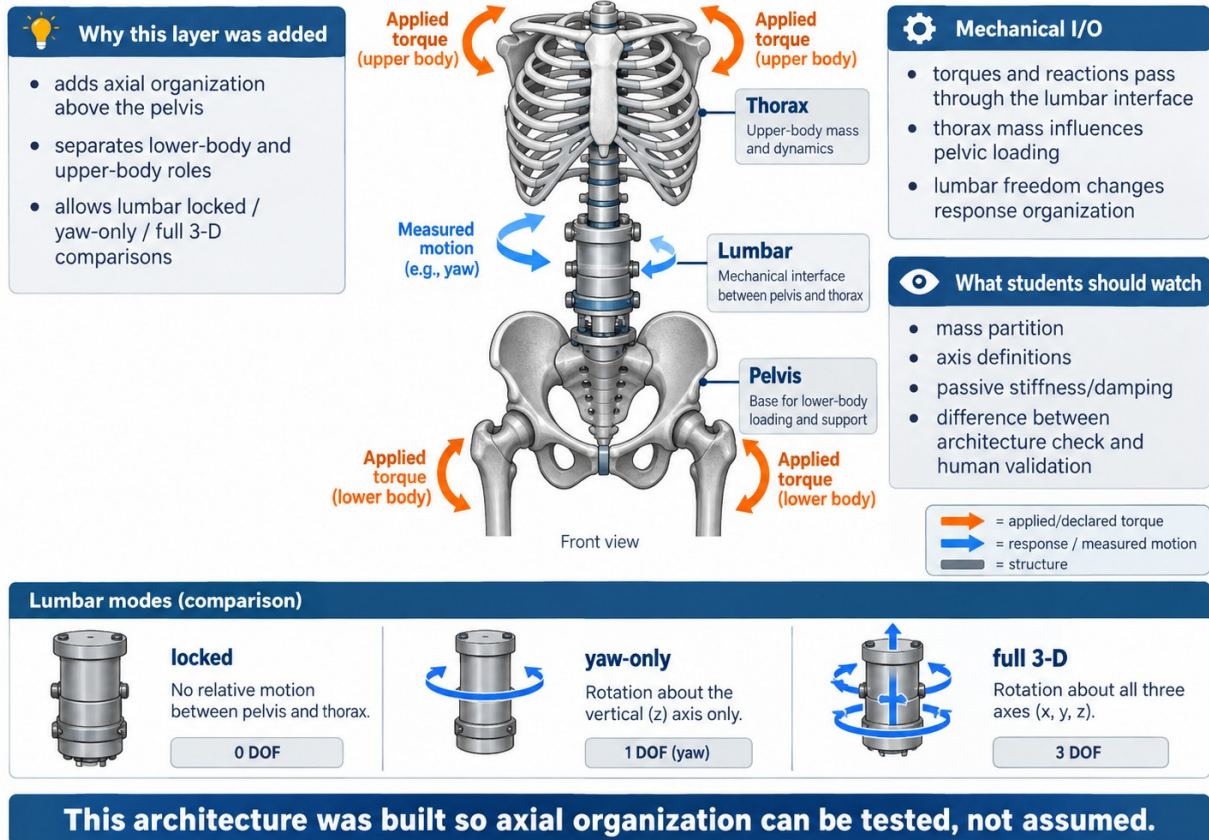
ModelicaHumanBodyParts v0.10.0 RC1

Master Teaching, Reproduction and Public Understanding Guide

ModelicaHumanBodyParts + THE LYNKORR PROJECT

For dancers, general readers and engineering students

Pelvis-lumbar-thorax architecture



The pelvis-lumbar-thorax layer is shown here as a teaching picture. The mechanical model is reduced and verified as software/mechanics; it is not an anatomical validation.

Publication candidate source SHA-256

2fda3b797ef9304529776d0610a3a8608e233c92531924579cb2eafe6665de4b

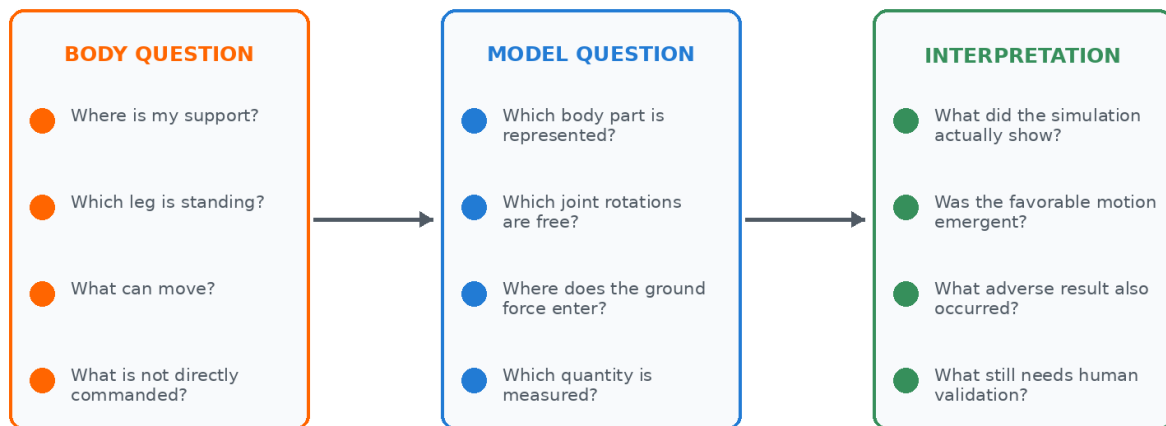
OpenModelica 1.27.1 (64-bit) | Modelica Standard Library 4.1.0 | Teaching edition: 24 September 2026

How to use this book

This guide deliberately has two reading levels. A dancer or general reader can understand the body question and the diagrams without reading Modelica code. An engineering student can continue into the equations, interfaces, verification tests and reproduction workflow.

How dancers can read this model without becoming programmer

Start with the body question. Open the equations only when you want the engineering detail.



The guide therefore has two layers: plain language first, engineering detail second.

Two reading tracks share the same scientific rule: first identify the body question, then the model boundary, then the measured quantity, and only then interpret the result.

WHAT IS IT?	WHAT DOES IT DO?	WHAT CAN WE CLAIM?
Name the physical or mathematical object before reading equations.	Separate what is commanded from what is allowed to emerge.	State what the test establishes - and what still needs human evidence.

The five questions to ask on every page

- What physical question is being asked?
- What is fixed, prescribed or commanded?
- What is free to move or emerge from the coupled equations?
- Which output actually answers the question?
- What can the result establish - and what does it still not establish?

WHAT THIS DOES NOT PROVE

A computer result can be internally correct and still be an unsuitable description of a real dancer. Verification and human validation are separate evidence layers.

1. What the v0.10 model is - and is not

ModelicaHumanBodyParts is a modular equation-based multibody library. It represents selected body segments as rigid bodies, joints as explicit rotational coordinates, foot-ground interaction as a reduced contact model, and selected measurements as forces, moments, power, energy and kinematic diagnostics.

ModelicaHumanBodyParts v0.10 - at a glance

A public map from human description to measured mechanics

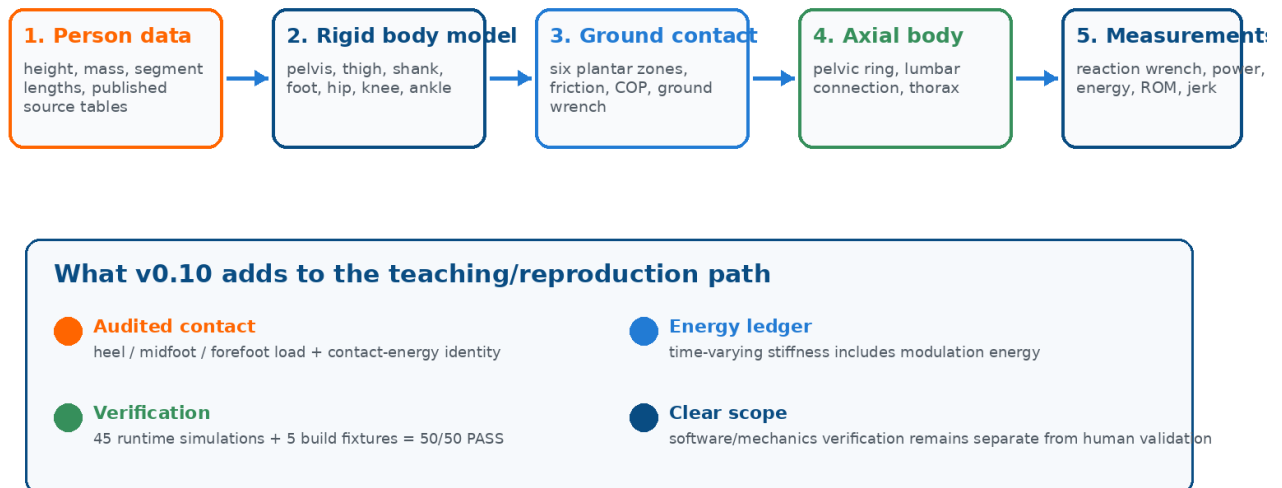


Figure 1. The v0.10 reading path from subject data to mechanical measurements.

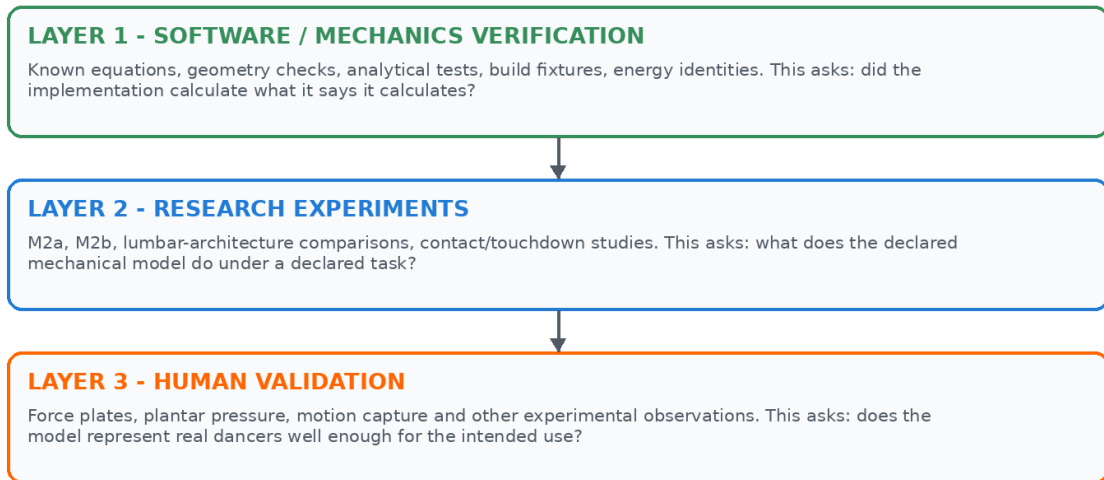
FOR DANCERS / GENERAL READERS

Think of the library as a transparent mechanical laboratory. It lets us ask questions such as: where is support carried, which joint is moving, where does a reaction force travel, how does the pelvis and trunk reorganize, and whether a free leg moves without being directly commanded. It does not decide whether a movement is beautiful or whether one technique is universally best.

Term	Plain-language meaning
Rigid segment	A mass with fixed internal distances, COM and inertia. The whole segment may translate and rotate.
Joint coordinate q	One relative rotation allowed by a joint. The axis and sign must always be named.
Mechanical frame	A connection that carries pose/velocity compatibility and a force/moment wrench.
Contact	A numerical law that creates ground reaction when a plantar point compresses the ground plane.
Observer / sensor	A calculation that measures something without necessarily creating a new physical force.
Verification	Evidence that code/equations reproduce a declared numerical contract.
Human validation	Evidence that the model matches real people well enough for the intended use.

Three evidence layers - do not mix them

A simulation can be correct as software without yet being validated as a human



v0.10 is strong in Layer 1, useful in Layer 2, and does not yet claim Layer 3.

Figure 2. The three evidence layers must remain separate throughout the project.

2. What changed in v0.10.0 RC1

v0.10 keeps the verified lower-body and axial mechanics while adding a publication-oriented audit layer around plantar support and time-varying mechanical impedance. The goal is not to hide the older models, but to make the active evidence path more explicit.

AUDITED CONTACT Contact.AuditedPlantarContact reports heel, midfoot and forefoot load, broad support, tangential travel and an energy identity.	ENERGY LEDGER Variable stiffness is treated as an explicit energy exchange. This prevents a changing coefficient from appearing as unexplained energy.	RELEASE VERIFICATION The publication candidate register contains 45 runtime simulations and 5 build-only fixtures: 50/50 PASS in the verified environment.
---	--	--

The older v0.9.2 release remains historically frozen. v0.10 is a separate publication candidate; old run folders and old evidence are not overwritten.

v0.10 verification - what 50/50 PASS actually means

Verification protects a declared software/mechanics contract. It is not a realism score.



PASS = the declared numerical contract was reproduced in the verified environment.

Figure 3. What the v0.10 release register means.

3. Coordinates, motion and load transmission

Before discussing tango or support, the coordinate convention has to be unambiguous. The library uses +x forward, +y left and +z upward. Long lower-limb segments point mainly toward -z in neutral stance.

Joint coordinates used by the library

The arrows show the positive motion directions in the library coordinate convention

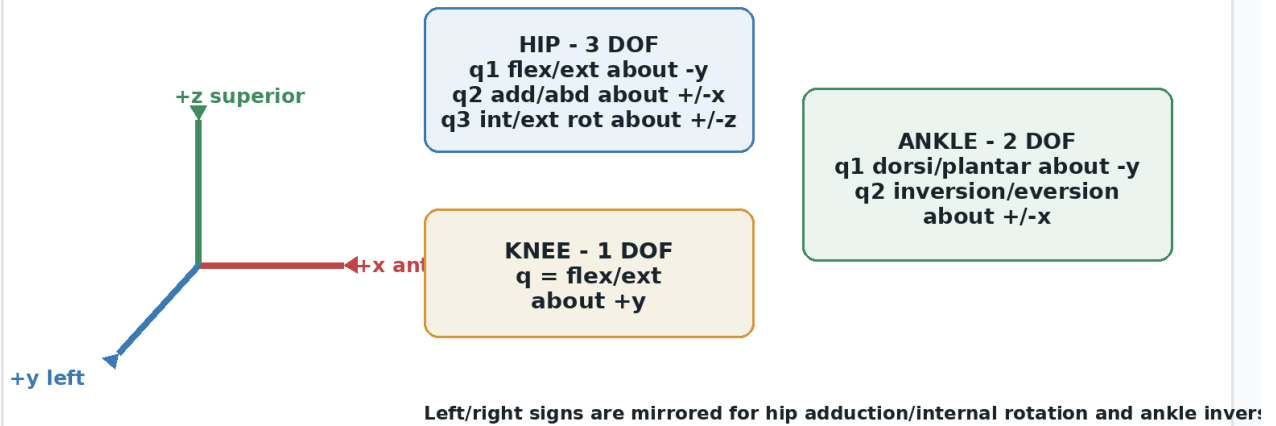
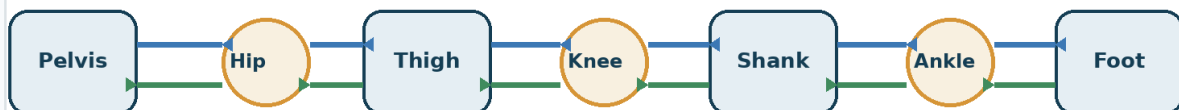


Figure 4. Joint coordinates and axis conventions used by the lower-body library.

How motion and load travel through one leg

Forward kinematics goes down the chain; ground reactions travel back up through the same connections

Motion / kinematics: position, angle, velocity, acceleration



Reaction loads: force and moment from contact / inertia

Rigid segment: no relative motion inside the part
Joint: allows only its declared rotational DOF(s); all other relative motions are constrained

Figure 5. Motion and reaction loads use the same connected mechanical chain. The solver satisfies the complete system simultaneously.

ENGINEERING LAYER

For a rigid body: $\Sigma F = m a_{COM}$ and $\Sigma M_{COM} = I \alpha + \omega \times (I \omega)$. Forces and moments must be expressed at a common point and in a common frame before they are added. A force arrow without a body, application point and coordinate frame is incomplete information.

$$\Sigma F = m a_{COM} \quad ; \quad \Sigma M_{COM} = I \alpha + \omega \times (I \omega)$$

PART I - THE BODY MODEL

These chapters explain the physical components first, using the same visual language as the deep teaching guide. Every section begins in plain language and then adds the engineering detail needed for reproducibility.

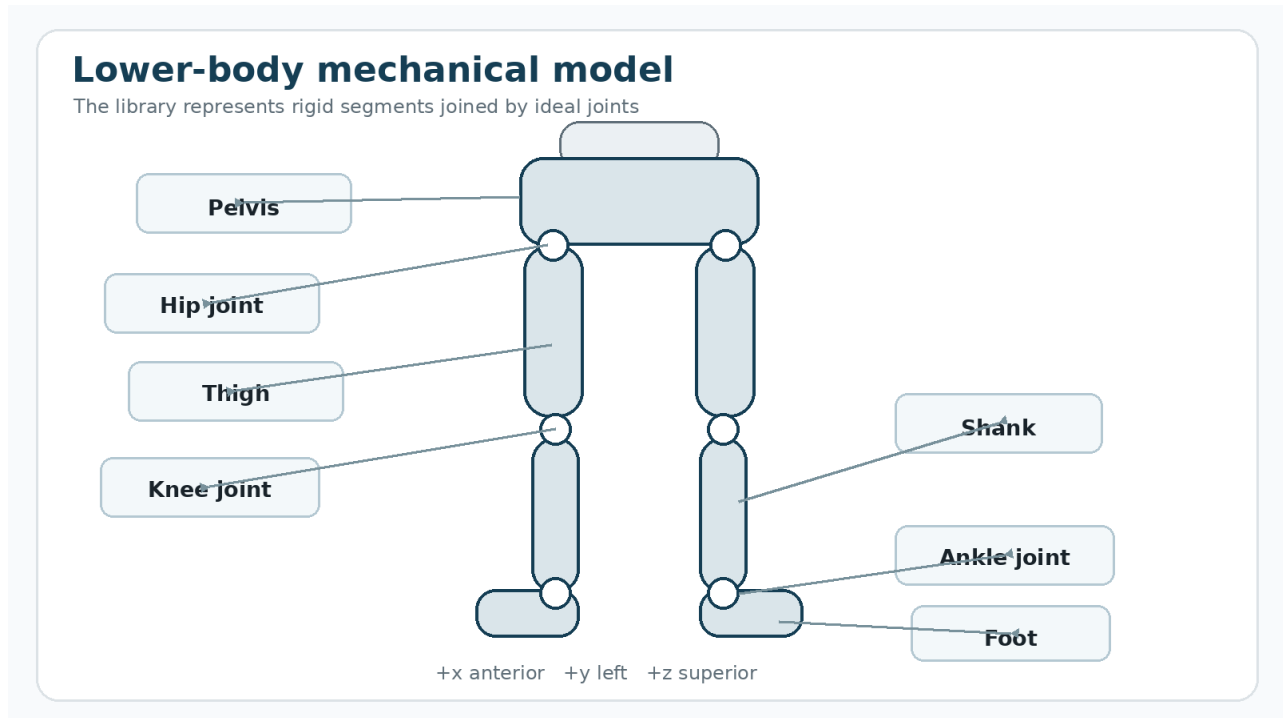
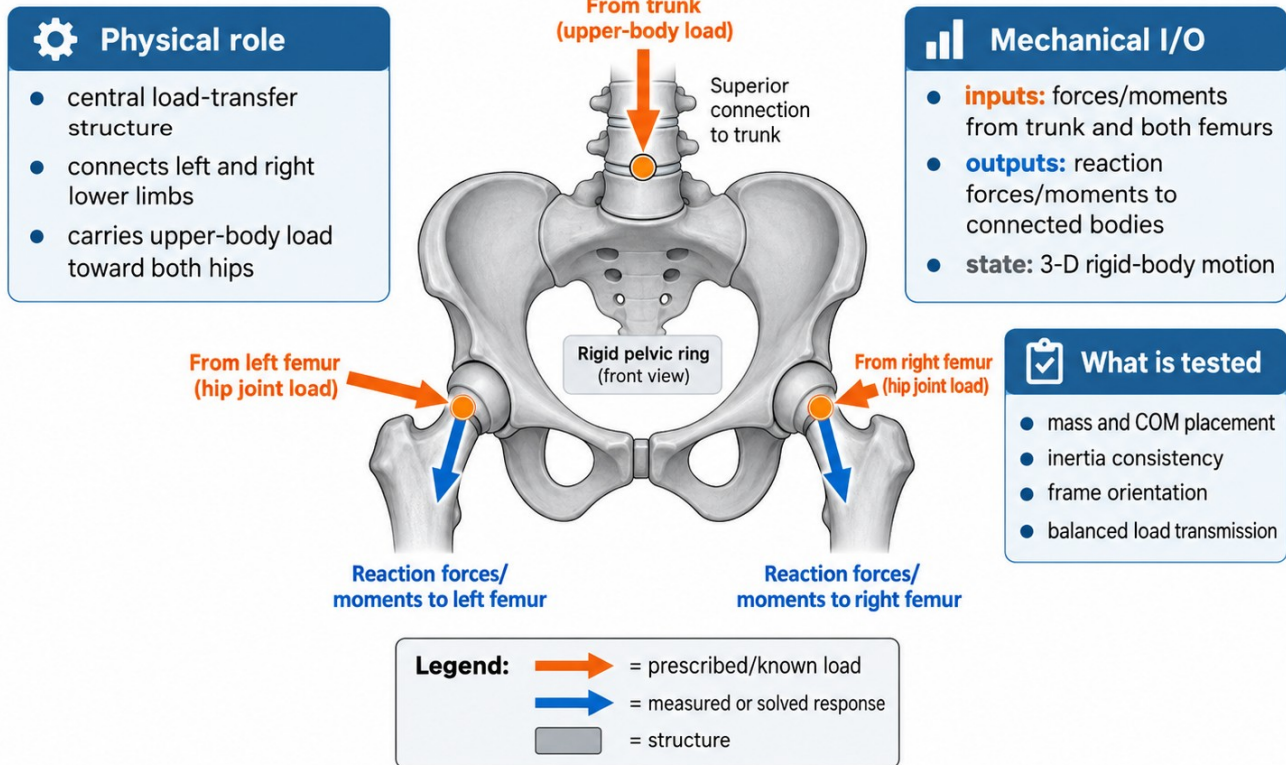


Figure 6. Lower-body mechanical model: rigid segments joined by ideal rotational joints.

4. Pelvis and pelvic ring

Pelvis component



Read the pelvis as a load-transfer hub, not as an isolated bone.

Figure 7. Read the pelvis as a load-transfer hub. In the actual library the legacy `Segments.Pelvis proxy` and the newer `AxialBody.PelvicRing` are distinct classes.

WHAT IS IT?	WHAT DOES IT DO?	WHY DANCERS CARE
A rigid central body that connects the two hip branches and carries load between lower body and trunk.	Maintains fixed geometry between its attachment frames while Newton-Euler equations determine its whole-body motion and reactions.	Standing-leg forces and moments reach the pelvis through the hip; pelvic motion then changes the mechanical boundary seen by the other leg.

FOR DANCERS / GENERAL READERS

In dance language, the pelvis is not modeled as a “command center.” It is a rigid mechanical hub. If it rotates or translates, that motion must come from the connected loads and joints, unless a boundary condition explicitly prescribes it.

Engineering item	v0.10 interpretation
Legacy class	<code>Segments.Pelvis</code> - reduced lower-trunk/pelvis proxy with fixed bilateral hip frames.
Current axial class	<code>AxialBody.PelvicRing</code> - rigid pelvis with left hip, right hip and lumbosacral connection.
Internal mobility	0 DOF inside the rigid pelvic body.
Mechanical I/O	Three-dimensional reactions at the connected frames; whole-body pose and acceleration are solved by the multibody system.

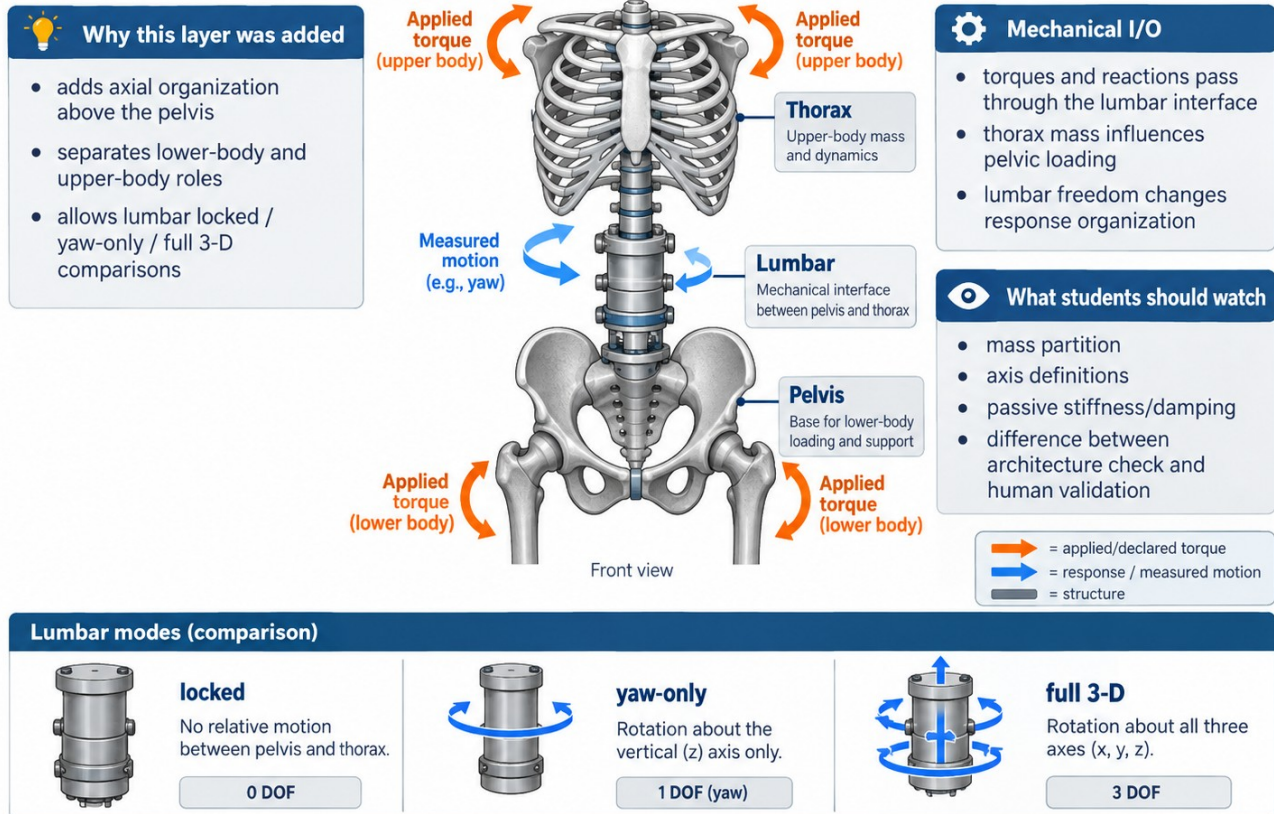
Verification	NeutralStanceGeometryTest, ScalingSmokeTest; PelvicRingGeometryTest and AxialMassPartitionTest for the axial body.
--------------	--

WHAT THIS DOES NOT PROVE

The pelvis components do not model sacroiliac deformation, pelvic-floor mechanics or a complete anatomical pelvis. The old lower-trunk proxy must not be added on top of new axial-body masses without a mass-partition audit.

5. Pelvis-lumbar-thorax architecture

Pelvis-lumbar-thorax architecture



This architecture was built so axial organization can be tested, not assumed.

Figure 8. The axial layer makes pelvis-to-thorax organization measurable instead of assuming a rigid trunk.

LOCKED	YAW ONLY	FULL 3-D
No relative lumbar motion. Pelvis and thorax behave as one rotational unit.	One relative axial-rotation coordinate between pelvis and thorax.	Flexion/extension, lateral bend and axial rotation are all available.

FOR DANCERS / GENERAL READERS

This matters because visible upper-body quietness does not imply that the pelvis and trunk are mechanically locked. The model lets us compare a rigid trunk with internal axial organization while keeping the same lower-body task.

ENGINEERING LAYER

AxialBody.LumbarSpine3D uses three serial revolute coordinates. The instrumented Transmission.LumbarSpine3D adds reaction wrench, generalized torque/power and variable impedance. In the controlled architecture study, the locked model produced a very different free-side response, while yaw-only and full 3-D were much closer for the tested predominantly rotational task.

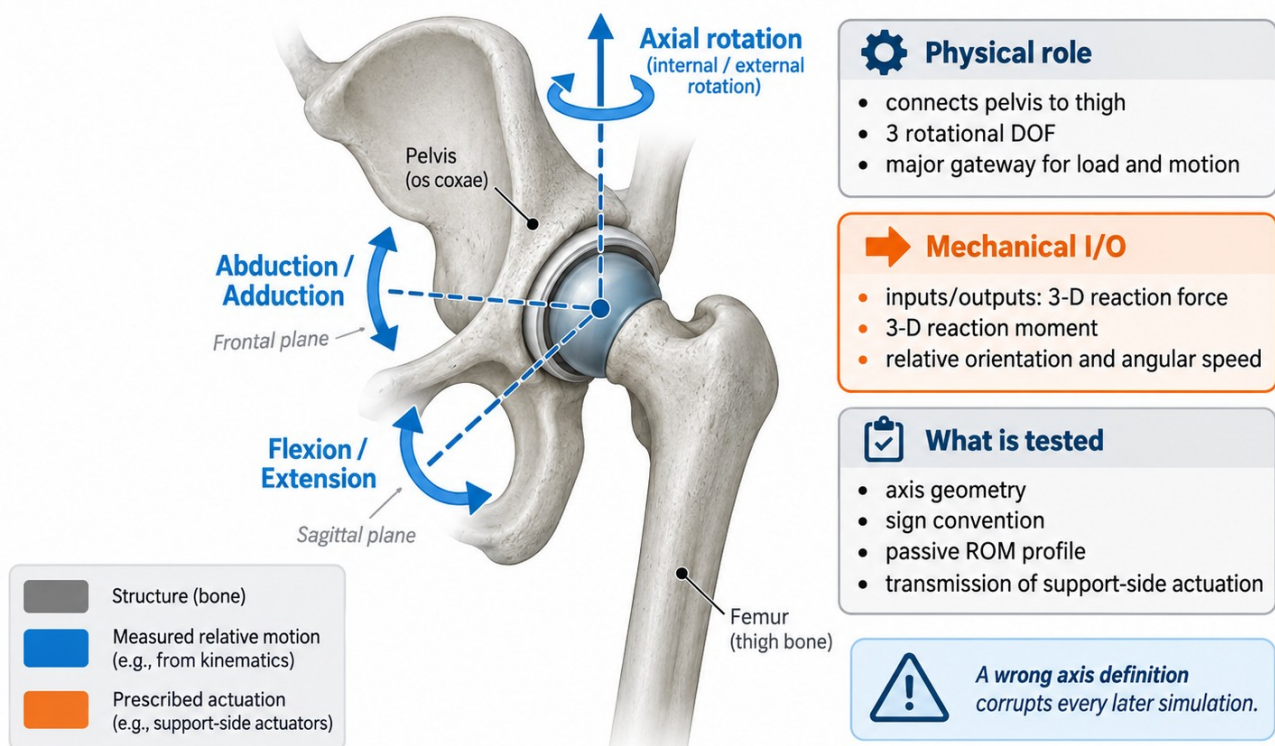
Class / item	Role
PelvicRing	Rigid pelvis with bilateral hip frames and lumbosacral frame.
LumbarSpine3D	3 rotational DOF: flex/extend, lateral bend, axial rotation.
Thorax	Rigid upper-trunk mass with lumbar, cervical and shoulder reference frames.
PelvisLumbarThorax	Reusable axial subassembly containing the three components.
Verification	PelvicRingGeometryTest, LumbarAxisGeometryTest, AxialMassPartitionTest, LumbarPassiveDissipationTest, rotational-transmission tests.

WHAT THIS DOES NOT PROVE

The lumbar model is a reduced 3-coordinate connection, not vertebra-by-vertebra anatomy. A software/model-reduction result is not a statement about individual spinal health or mobility.

6. Hip joint

Hip joint component



Always name the axis and the sign of the reported hip quantity.

Figure 9. The reduced hip has three rotational coordinates and carries reaction loads between pelvis and thigh.

FOR DANCERS / GENERAL READERS

The hip is one of the key bridges between the standing leg, pelvis and free leg. In the model, “opening” must be stated as a particular coordinate or endpoint motion - not as a vague visual impression.

Engineering item	Meaning
Coordinates	q1 flexion/extension; q2 adduction/abduction; q3 internal/external rotation.
Left/right signs	The second and third axes are mirrored so the same clinical direction has a consistent interpretation on each side.
Legacy Joints.Hip	Ideal 3-DOF joint with ROM monitoring and optional fixed passive spring-damper.
Transmission.Hip	Adds variable impedance, reaction force/moment, generalized torque/power and cut-power closure.
Verification	HipAxisGeometryTest, NeutralStanceGeometryTest, JointLimitMonitorTest; transmission analytical tests protect power/impedance bookkeeping.

$$P_{\text{joint}} = \tau_1 \dot{q}_1 + \tau_2 \dot{q}_2 + \tau_3 \dot{q}_3$$

WHAT THIS DOES NOT PROVE

The model does not identify muscles, joint contact stress, cartilage load or ligament tension. A 3-DOF ideal serial joint is a system-mechanics reduction.

7. Thigh and shank rigid segments

THIGH	SHANK	KEY IDEA
Rigid hip-to-knee segment carrying mass, COM and inertia.	Rigid knee-to-ankle segment using the same long-segment base mechanics.	The joints create mobility; the segments carry geometry, inertia, force and moment between joints.

Both Segments.Thigh and Segments.Shank extend Base.PartialLongSegment. Inside each rigid segment, the proximal-to-distal distance is fixed. The whole segment can still translate and rotate because its neighboring joints move.

$$\mathbf{r}_{\text{distal}} = \{0, 0, -L\} \quad ; \quad \Sigma \mathbf{F} = m \mathbf{a}_{\text{COM}} \quad ; \quad \Sigma \mathbf{M}_{\text{COM}} = \mathbf{I} \boldsymbol{\alpha} + \boldsymbol{\omega} \times (\mathbf{I} \boldsymbol{\omega})$$

ENGINEERING LAYER

Base.RigidSegment3D is the more general route when a full 3-D COM vector and complete inertia tensor are already known. The scalar long-segment route remains useful because its assumptions are transparent and source-traceable.

WHAT THIS DOES NOT PROVE

Rigid thigh/shank components do not contain muscle, tendon, bone deformation or soft-tissue wobble.

8. Knee joint

Knee joint component


Secondary motions (limited)

- small internal/external rotation
- small varus/valgus (frontal plane)


Physical role


- connects thigh and shank
- mainly flexion/extension
- transmits large stance loads

Mechanical I/O


Reaction force $\vec{F}$
(measured/solved output)



Reaction moment $\vec{M}$
(measured/solved output)



Relative angle and angular speed $(\theta, \dot{\theta})$
(measured/solved output)

What is tested


- axis geometry
- ROM limits
- sign conventions
- load transmission under stance

Legend


Prescribed input
(e.g., motion, load, boundary condition)



Measured/solved output
(e.g., reaction, kinematics, kinetics)



Structure
(e.g., bones, joint component)


Student caution:

Do not interpret every reported knee moment as active muscle effort.

The knee is both a motion joint and a force-transmission location.

Figure 10. The knee is both a motion coordinate and a force-transmission location.

FOR DANCERS / GENERAL READERS

A small visible knee angle can still coexist with a large reaction moment. This is why the guide keeps “how much it moves” separate from “how much load it transmits.”

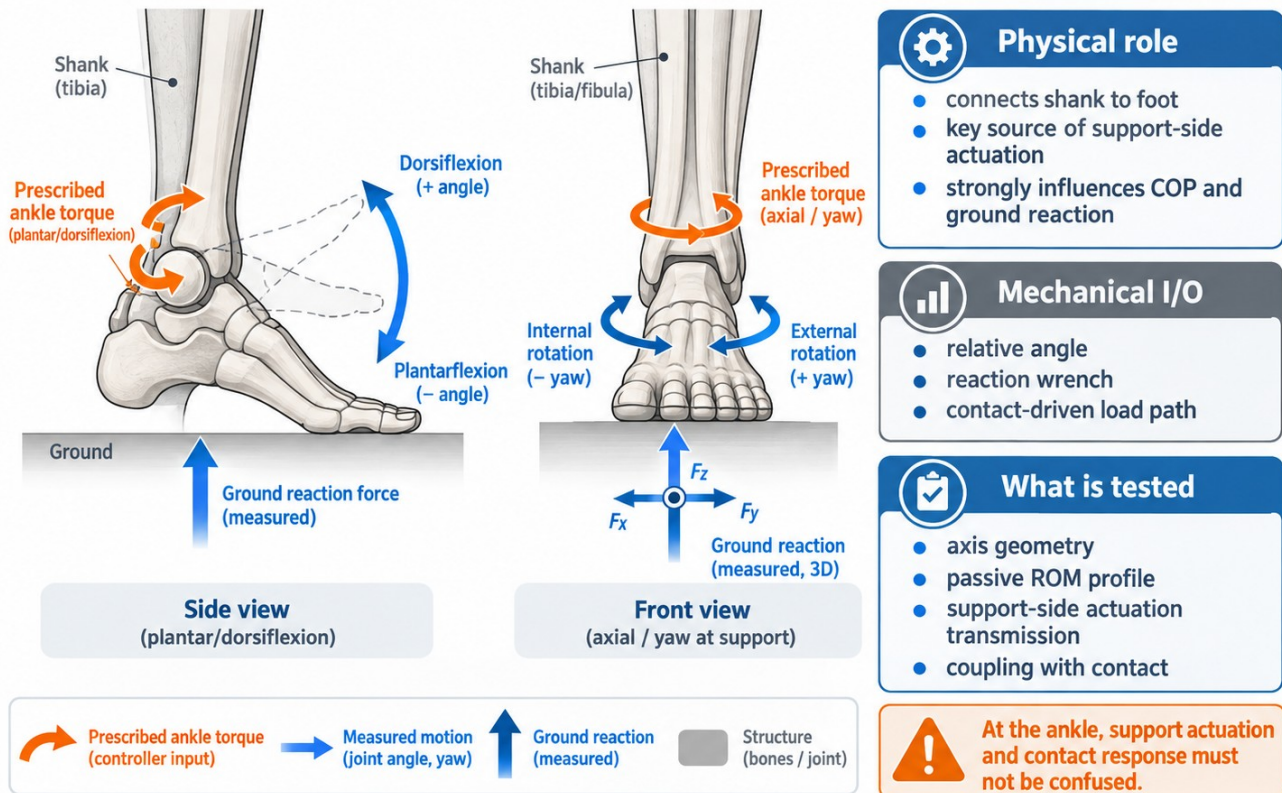
Item	Meaning
Mobility	1 rotational DOF: flexion/extension about the declared +y axis.
Joints.Knee	Reports angle and ROM diagnostics.
Transmission.Knee	Adds variable impedance, reaction wrench, generalized torque/power and cut-power closure.
ROM monitor	Reports distance/utilization relative to the declared range; it does not mechanically stop the joint.
Verification	KneeAxisGeometryTest, NeutralStanceGeometryTest, JointLimitMonitorTest.

WHAT THIS DOES NOT PROVE

No rolling/sliding arthrokinematics, patella, ligaments or tissue stress are modeled.

9. AnkleComplex

AnkleComplex component



In Lynkorr questions, the ankle is often the first active driver on the support side.

Figure 11. The reduced ankle provides the two foot-orientation coordinates used immediately above the ground-contact layer.

q1 Dorsiflexion / plantarflexion.	q2 Inversion / eversion with mirrored left/right sign.	MEASUREMENTS Angle, speed, reaction wrench, power and ROM diagnostics depend on which wrapper is used.
---	--	--

FOR DANCERS / GENERAL READERS

The ankle is the first articulated joint above the foot-ground interface. It can therefore change how the rigid foot presents its six contact locations to the floor, but it is not a complete anatomical ankle/subtalar model.

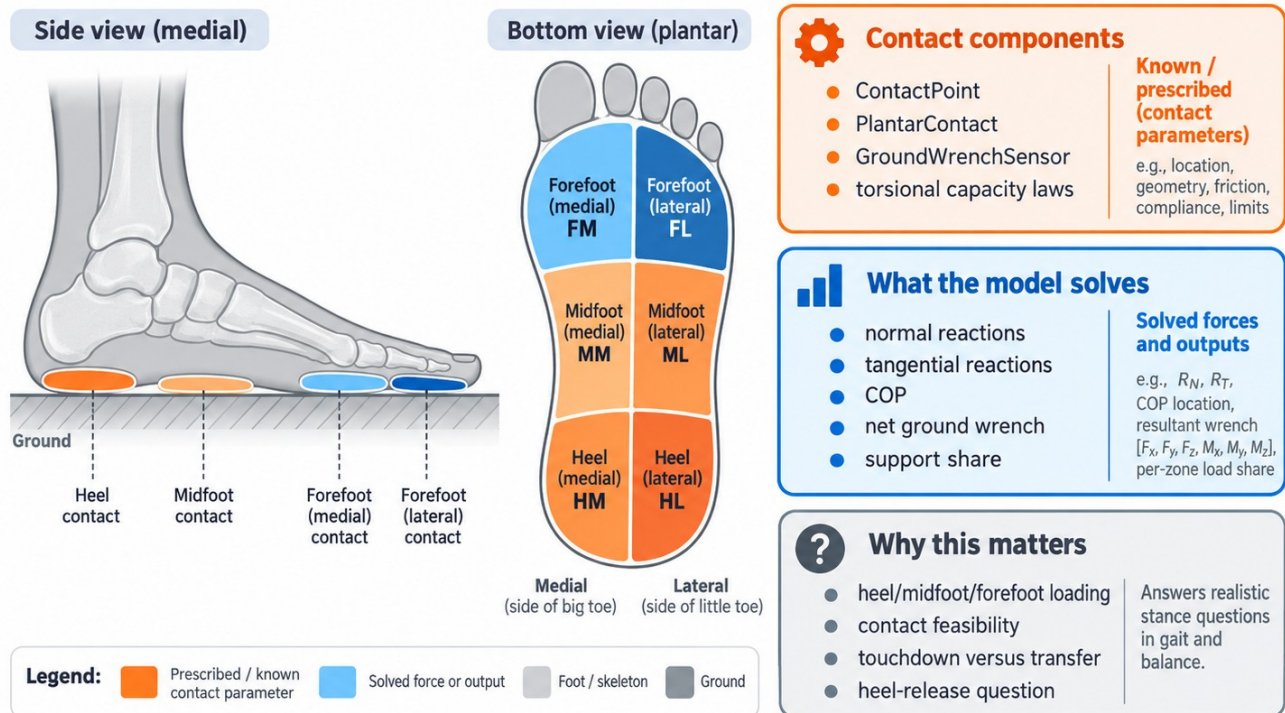
WHAT THIS DOES NOT PROVE

The current 2-DOF abstraction does not claim physiological talocrural/subtalar decomposition or a universal ankle range.

10. Foot and six-zone contact stack

Foot-ground contact stack

Six plantar contact zones, model components, outputs, and why it matters.



Realistic stance questions require explicit plantar contact, not only an ideal fixed support.

Figure 12. The foot is a rigid body; the contact system underneath it is a separate set of numerical contact elements and observers.

Segments.Foot defines heel, midfoot, toe, medial/lateral landmarks and six plantar frames. Those frames can be connected to Contact.PlantarContact or Contact.AuditedPlantarContact. The foot itself does not decide the contact force; the contact law does.

ENGINEERING LAYER

The rigid foot transmits the sum of distributed contact forces and moments back through the ankle. Six numerical plantar zones are a discretization for force routing; they are not six anatomical pressure sensors.

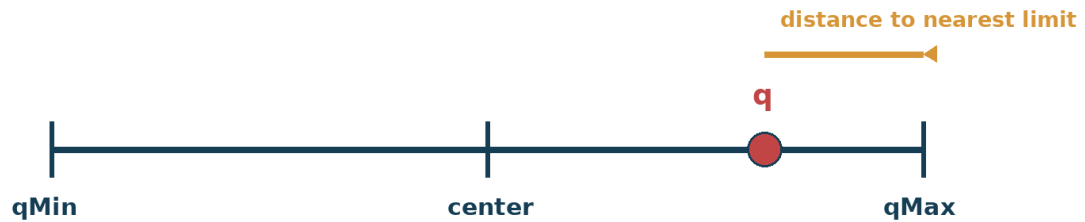
WHAT THIS DOES NOT PROVE

The default foot COM/zone geometry and numerical contact parameters are modeling assumptions unless separately calibrated.

11. Range-of-motion monitoring

Joint range-of-motion monitor

It measures how close a joint coordinate is to its declared limit



The monitor reports the limit; it does NOT physically stop the joint.

Figure 13. JointLimitMonitor reports distance to a declared mobility envelope; it does not create a physical stop.

FOR DANCERS / GENERAL READERS

A dancer can move close to a ROM boundary without that boundary being the mechanical cause of the movement. The monitor answers “how close?” - not “what physically stopped the body?”

$$\text{marginLower} = q - q_{\text{Min}} \quad ; \quad \text{marginUpper} = q_{\text{Max}} - q$$

WHAT THIS DOES NOT PROVE

Published or assumed ROM values are not universal limits and are not individualized without subject-specific evidence.

PART II - GROUND CONTACT AND SUPPORT

This part explains how a planted or moving foot exchanges force and moment with the floor. It is central to Lynkorr questions because “support” has to be measured mechanically, not inferred from appearance.

12. One ContactPoint

One ground contact point

A simple numerical spring-damper + friction model

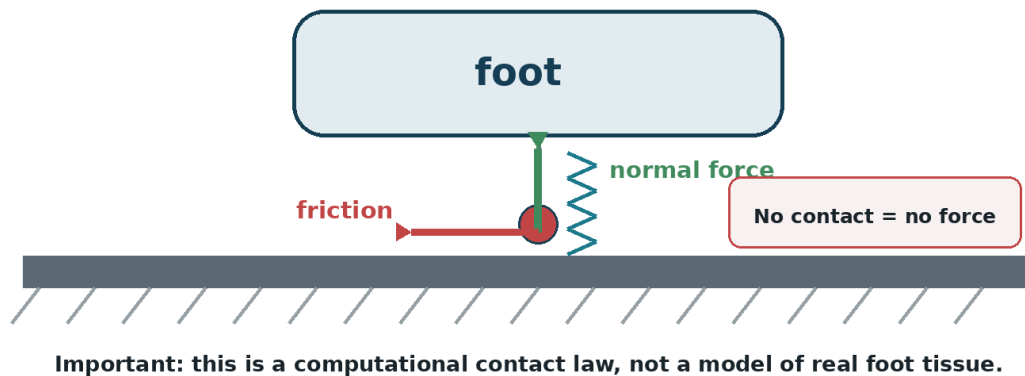


Figure 14. One ground contact point: compression creates a normal force; tangential motion creates a regularized friction force.

ENGINEERING LAYER

Contact.ContactPoint is unilateral: it can push but cannot pull the foot toward the floor. Normal stiffness and damping are computational contact parameters. Friction uses a smooth tanh regularization and therefore does not provide exact static sticking at zero slip.

$$\delta = \max(0, z_{\text{ground}} - z)$$

$$N = \max(0, k_{\text{normal}} \delta + c_{\text{normal}} \dot{\delta}) \quad \text{while } \delta > 0$$

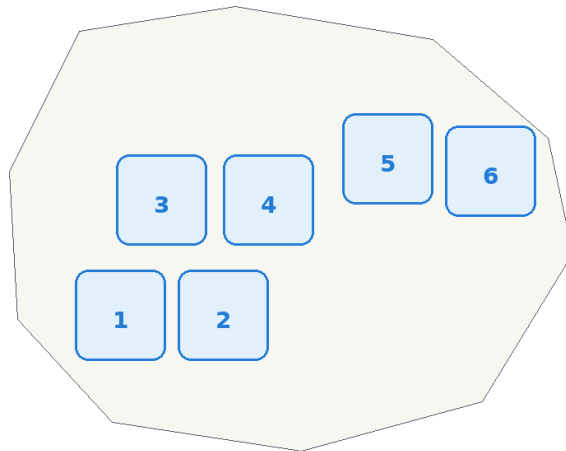
WHAT THIS DOES NOT PROVE

This is not a plantar-tissue, shoe or dance-floor constitutive model. Its parameters require calibration before physiological interpretation.

13. PlantarContact and AuditedPlantarContact

v0.10 audited plantar contact

The foot is not one support point: six zones carry and report load



Zones 1-2 heel | 3-4 midfoot | 5-6 forefoot

Regional support outputs

$$N_H = N1 + N2$$

$$N_M = N3 + N4$$

$$N_F = N5 + N6$$

$$\text{heelShare} = N_H / (N_H + N_M + N_F)$$

Energy audit

$$dU/dt + \text{lossPower} + \text{powerToBody} = 0$$

The wrapper also reports loaded tangential travel, supportValid and broadSupport.

Important: heel share is a load ratio, not anatomical contact area.

Figure 15. v0.10 adds regional support and contact-energy outputs without replacing the six-zone contact mechanics.

PlantarContact	AuditedPlantarContact	Dancer interpretation
Six ContactPoint instances + optional residual spin + GroundWrenchSensor.	Adds regional load, broad-support status, tangential travel and contact-energy identity.	Heel retention, heelward redistribution and broad support are different outcomes.

FOR DANCERS / GENERAL READERS

A rising heel percentage does not automatically mean “better connection.” It can rise because the forefoot has unloaded. Therefore the guide reports absolute heel force and whether midfoot/forefoot remain loaded, not only a percentage.

WHAT THIS DOES NOT PROVE

Six loaded numerical zones do not equal biological contact area. COP does not uniquely determine regional pressure.

14. Resultant ground force, moment and COP

Ground reaction force, moment and COP

GroundWrenchSensor combines all contact loads into one equivalent wrench

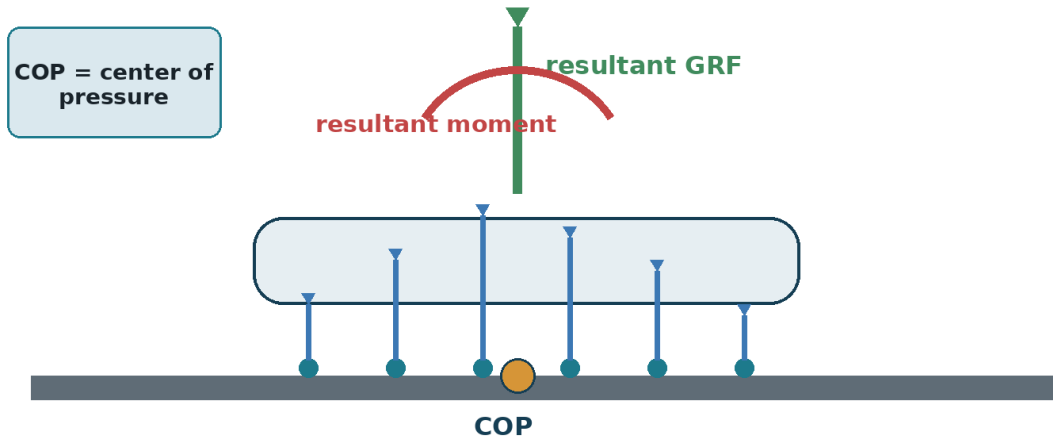


Figure 16. GroundWrenchSensor converts local forces into the equivalent force-plate quantities.

$$F = \sum F_i \quad ; \quad M_O = \sum [(r_i - O) \times F_i + M_{\text{free},i}]$$

The center of pressure (COP) is a wrench-equivalent point on the declared reporting plane. It is not automatically the same as the normal-pressure centroid, especially when shear forces, free moments or a nonzero reporting height are involved.

ENGINEERING LAYER

GroundWrenchSensor also separates yaw produced by shear moment arms from an explicitly supplied free/torsional contact moment. This prevents a normal-pressure shift from being mistaken for yaw torque.

WHAT THIS DOES NOT PROVE

A correct wrench calculation can summarize unrealistic contact forces perfectly. Sensor verification does not validate the upstream contact law.

15. Friction and torsional spin

Two separate ideas are preserved in the library. The older torsionalCapacity function is retained for regression/history. The active TorsionalContactLaw uses residual local friction capacity so translational shear and spin do not spend the same Coulomb budget twice.

$$\text{remaining}_i = \max(0, \mu N_i - |F_{t,i}|)$$

$$C_i = \text{reserveFraction} \times (2 a_i / 3) \times \text{remaining}_i$$

FOR DANCERS / GENERAL READERS

Rotating "against the ground" requires a tangential/torsional ground action. Moving the COP by vertical pressure alone is not the same thing as creating a yaw moment.

WHAT THIS DOES NOT PROVE

The residual spin model is a conservative reduced contact assumption, not an experimentally identified footwear or plantar torsion law.

PART III - DATA, BODY SIZE AND MASS PROPERTIES

Before a body can move in simulation, its segment lengths, masses, centers of mass and inertias must be defined. This layer is where scientific source data and explicit modeling assumptions enter.

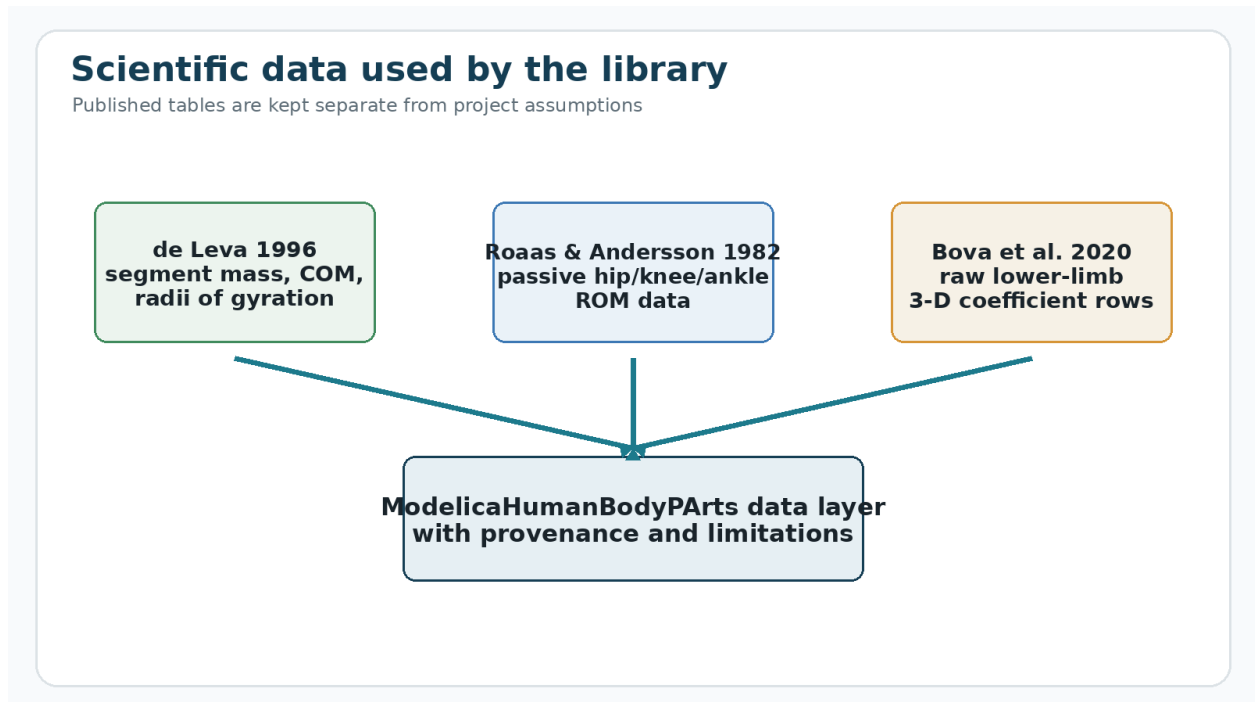


Figure 17. Scientific data sources are stored separately from project assumptions.

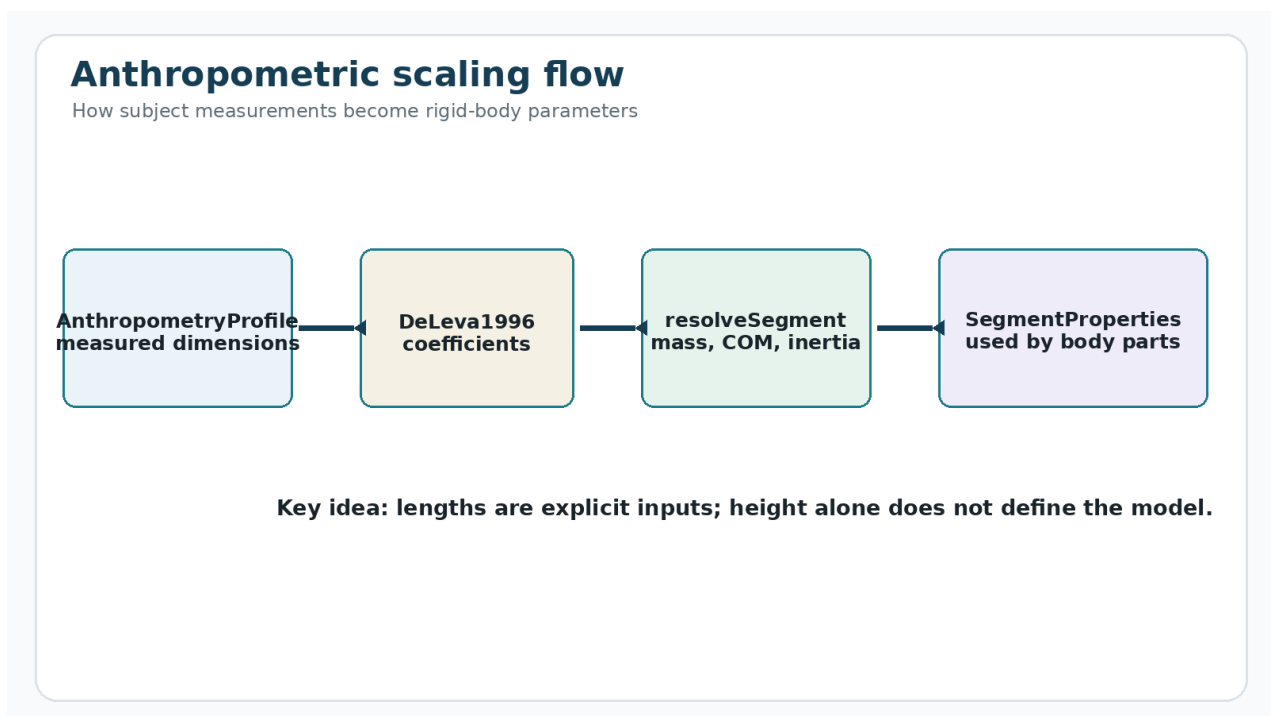


Figure 18. Anthropometric scaling path from subject measurements to segment mechanical properties.

AnthropometryProfile Stores the declared person/geometry inputs.	Source tables de Leva, Roaas-Andersson and Bova rows are kept with provenance and limits.	resolveSegment Combines explicit segment length + body mass fraction + source coefficients to create SegmentProperties.
--	---	---

$$m_i = f_i M \quad ; \quad c_i = cFrac_i L_i \quad ; \quad I_{axis} = m_i (k_{axis} L_i)^2$$

FOR DANCERS / GENERAL READERS

Changing body size changes the mechanics because mass, COM distance and inertia change. But the current lower-body scaler uses explicit segment lengths; height alone does not magically create a complete personalized body.

3-D mass properties

Center of mass and inertia tensor describe how a rigid segment resists rotation

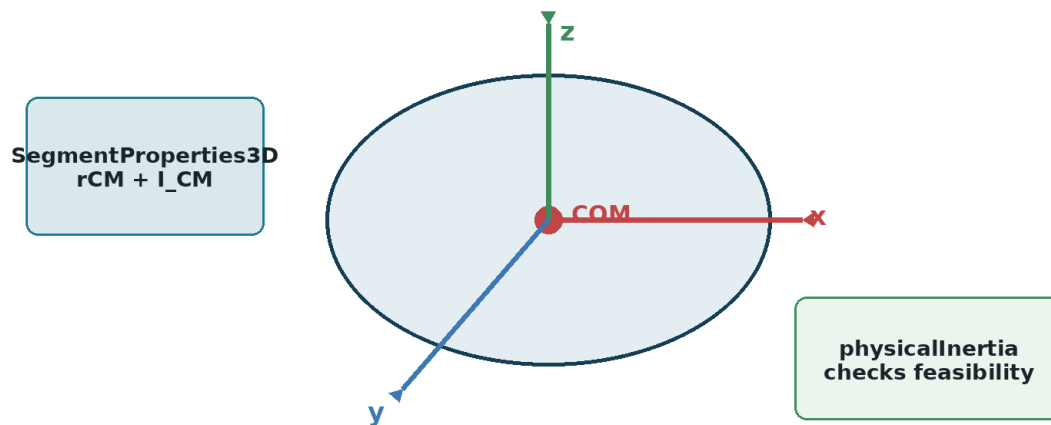


Figure 19. SegmentProperties3D can carry a full COM vector and inertia tensor when higher-fidelity data are available.

WHAT THIS DOES NOT PROVE

A source coefficient can be transcribed correctly and still be unsuitable if segment endpoints or coordinate frames are mismatched. Verification of a table is not validation of a person.

16. Behind-the-scenes data components

Component	Plain-language role
Types.Sex	Dataset selector for source rows; not a biological model of an individual.
Types.SegmentKey	Compiler-safe selector for Foot, Shank, Thigh or Pelvis in the active scalar scaler.
Types.ProvenanceClass	Separates published source, project derivation, explicit assumption and canonical experiment.
Types.ScalingLevel	Metadata describing how dimensions were obtained; does not silently switch algorithms.
Types.ROMNature	Labels mobility evidence such as passive/active/functional.
Records.BSIPCoefficients	Dimensionless mass/COM/radius coefficients used by the scalar scaler.
Records.SegmentProperties	Resolved scalar mass, length, COM and principal inertias.
Records.SegmentProperties3D	Full COM vector and full inertia tensor in a named segment frame.
Records.PublishedBSIPRow	Raw source-table row preserved before risky tensor conversion.
Utilities.physicalInertia	Checks whether an inertia tensor is physically realizable.

Utilities.affineMassProperties	Maps COM/inertia through a declared affine geometric transformation.
Utilities.shiftInertia	Parallel-axis / coordinate-rotation utility for inertia tensors.

PART IV - ASSEMBLIES, REACTION LOADS AND ENERGY

Individual components become useful only when they are connected into an experiment. This part explains the main assemblies and the instrumented Transmission layer used by the current research.

17. LowerBodyAssembly

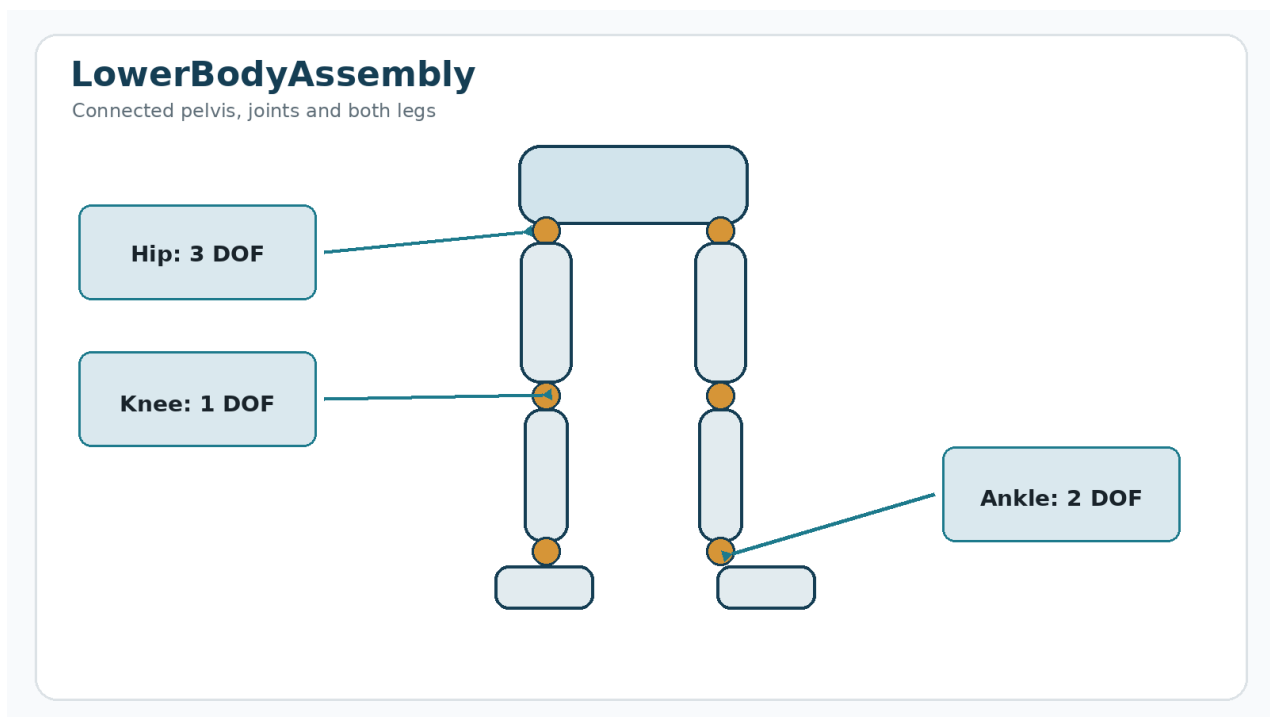


Figure 20. LowerBodyAssembly connects pelvis, both hips, thighs, knees, shanks, ankles and feet without ground contact.

Per leg the legacy articulated chain contains 3 hip + 1 knee + 2 ankle coordinates. The assembly exposes the plantar frames so contact can be attached externally. It requires an external boundary condition at the pelvis/root.

WHAT THIS DOES NOT PROVE

A connected assembly that builds successfully is not automatically a stable standing human. Boundary conditions matter.

18. LowerBodyGroundedAssembly

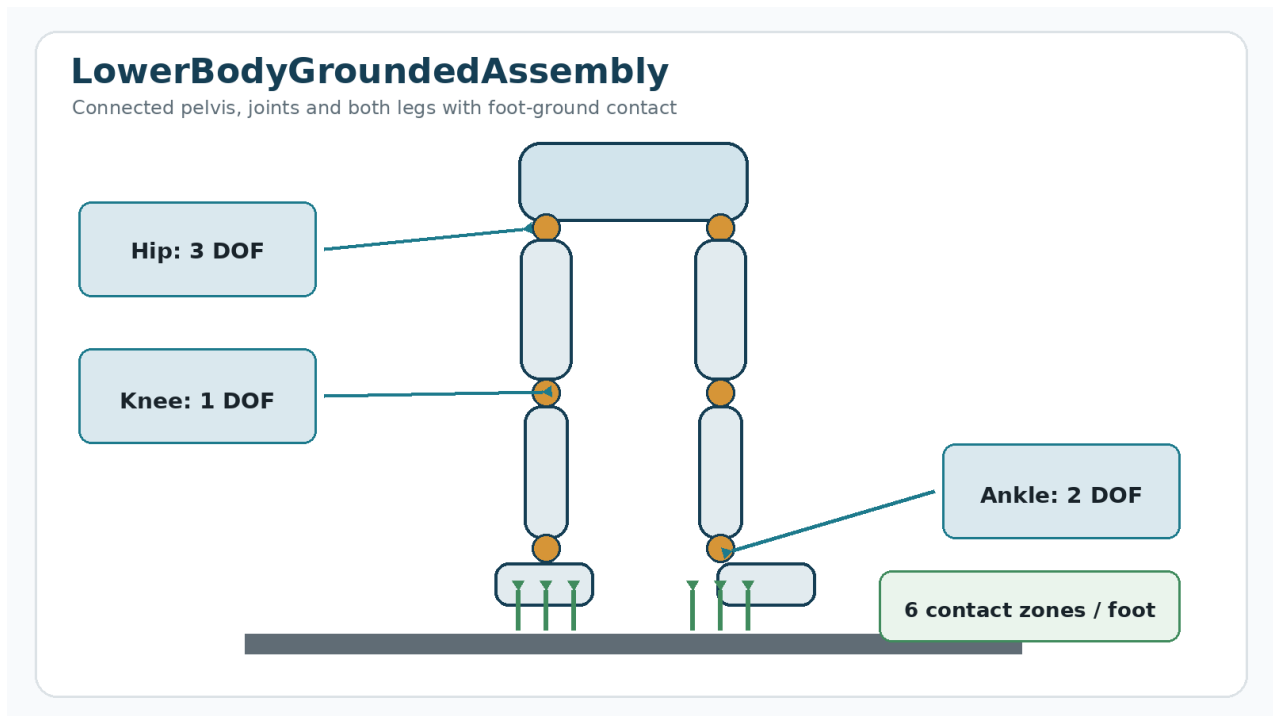


Figure 21. LowerBodyGroundedAssembly attaches a six-zone contact subsystem to each foot.

FOR DANCERS / GENERAL READERS

This is the cleanest educational picture of the external load path: floor → plantar zones → foot → ankle → shank → knee → thigh → hip → pelvis. The reaction loads and the kinematics are solved together.

19. Axial and Transmission assemblies

AxialBody.LowerBodyAxialAssembly replaces the old one-body lower-trunk proxy with PelvicRing + LumbarSpine3D + Thorax while keeping both articulated legs. AxialBody.LowerBodyAxialGroundedAssembly adds bilateral PlantarContact. Transmission.LowerBodyAxialAssembly then adds variable impedance, reaction wrench and power outputs at the major interfaces.

Class	Why it exists
AxialBody.PelvisLumbarThorax	Reusable pelvis-lumbar-thorax subassembly.
AxialBody.LowerBodyAxialAssembly	Axial body + both articulated legs.
AxialBody.LowerBodyAxialGroundedAssembly	Adds bilateral foot-ground contact.
Transmission.Knee / AnkleComplex / Hip / LumbarSpine3D	Instrumented wrappers with variable impedance, reaction wrench and power outputs.
Transmission.LowerBodyAxialAssembly	Whole instrumented body path with COM and seven cut-wrench channels.

20. Power, work and “load” are different quantities

Power and energy channels are different

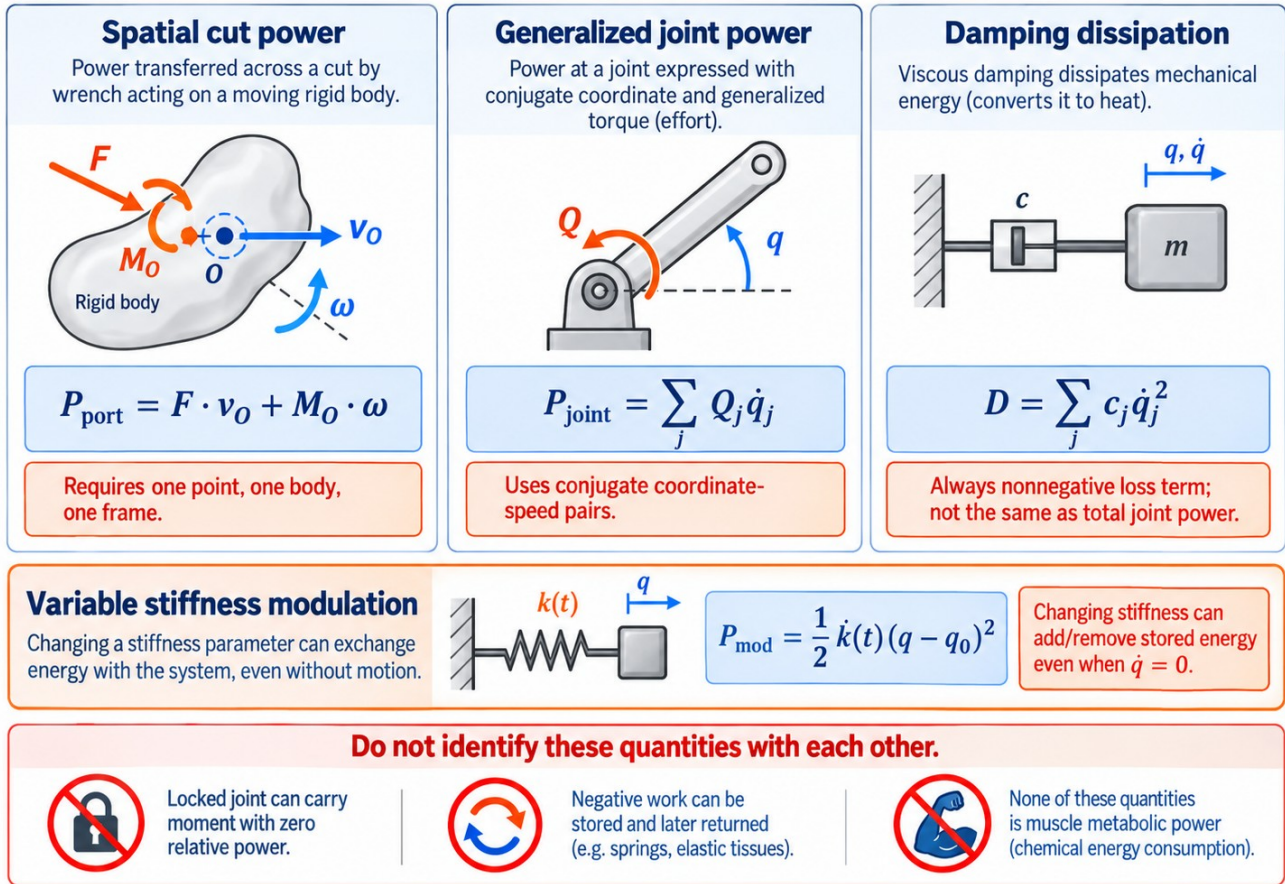


Figure 22. Spatial port power, generalized joint power, damping dissipation and stiffness-modulation power are related but not interchangeable.

$$P_{\text{port}} = F \cdot v + M \cdot \omega \quad ; \quad P_{\text{joint}} = \sum Q_j \dot{q}_j \quad ; \quad D = \sum c_j \dot{q}_j^2$$

FOR DANCERS / GENERAL READERS

A large moment with very little relative motion is not the same thing as large joint work. Likewise, a joint can move smoothly while an interface still carries a substantial reaction force. This is why the project records several metrics instead of declaring one “grounding score.”

WHAT THIS DOES NOT PROVE

These are mechanical quantities. They are not muscle metabolic power, pain, injury risk or tissue stress.

21. InterfaceLoadMetrics and KinematicSmoothnessMetrics

Observer	Outputs
InterfaceLoadMetrics	Force/moment norms, normalization by declared/represented weight, generalized power, loading rates, magnitude impulses, positive/negative work and a low-motion/high-load diagnostic.
KinematicSmoothnessMetrics	Acceleration magnitude, jerk vector/magnitude and accumulated jerk exposure.
SpatialWrenchPower	Combines a wrench and twist at a common point/frame to compute spatial mechanical power.

ENGINEERING LAYER

The low-motion/high-load ratio $B = |M|/(|\omega| + \epsilon)$ is an engineering diagnostic whose numerical value depends on the regularization ϵ and the chosen coordinate definition. It must not be presented as a biological injury index.

22. Variable impedance and its energy ledger

Transmission.VariableImpedance1D represents a rotational spring-damper whose stiffness and damping scales may vary in time. v0.10 protects the energy accounting by making stiffness-modulation power explicit.

$$\tau = -k(t)(q-q_0) - c(t) \dot{q}$$

$$U = 1/2 k(t)(q-q_0)^2$$

$$P_{\text{mod}} = 1/2 \dot{k}(t)(q-q_0)^2 \quad (\text{constant } q_0)$$

ENGINEERING LAYER

Changing stiffness can add or remove elastic storage even when ordinary torque-times-speed power is small. The element therefore reports modulationPowerIntoStorage separately and checks its energyBalanceResidual.

FOR DANCERS / GENERAL READERS

In Lynkorr studies this element may be used as a mechanical “release/holding” surrogate. It does not prove that breathing or muscle activation follows the chosen coefficient schedule.

PART V - HOW THE LIBRARY IS USED TO ASK LYNKORR QUESTIONS

The core library is reusable. The research experiments are separate fixtures built on top of it. This separation is important because a dancer-facing interpretation should never be confused with the software-verification layer.

23. M2a and M2b are different experiments

M2a vs M2b: two different experiments

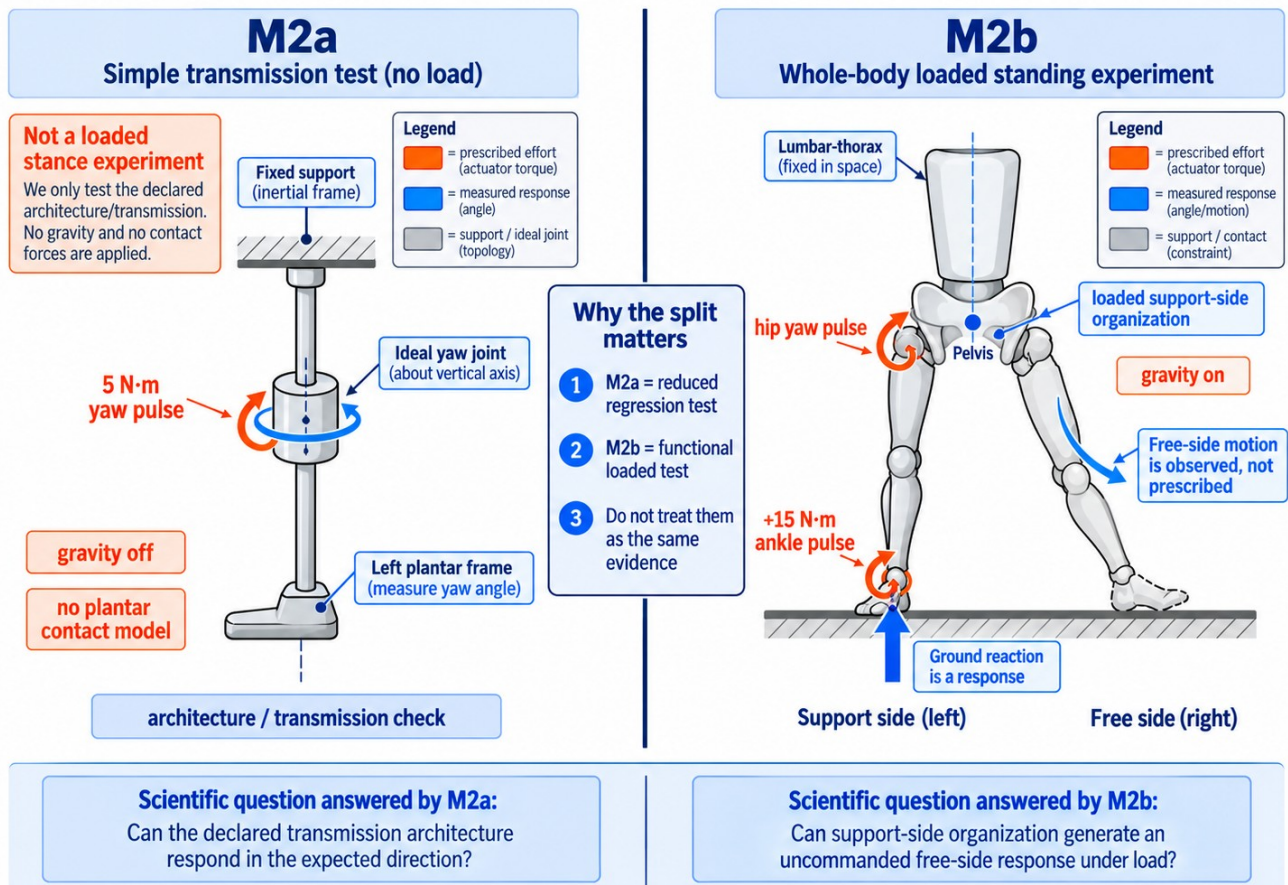


Figure 23. M2a is a transmission/architecture check; M2b is a gravity-loaded support-side experiment. A PASS in one is not a result for the other.

FOR DANCERS / GENERAL READERS

M2a asks “does a known rotational input transmit through the declared architecture?” M2b asks “under gravity, can support-side actuation produce an uncommanded response of the opposite side?”

WHAT THIS DOES NOT PROVE

Neither experiment by itself proves a general human Lynkorr mechanism. M2a is not loaded stance; the original core M2b fixture uses a fixed support boundary rather than a complete plantar transfer sequence.

24. Support-generated response

M2b support-generated response

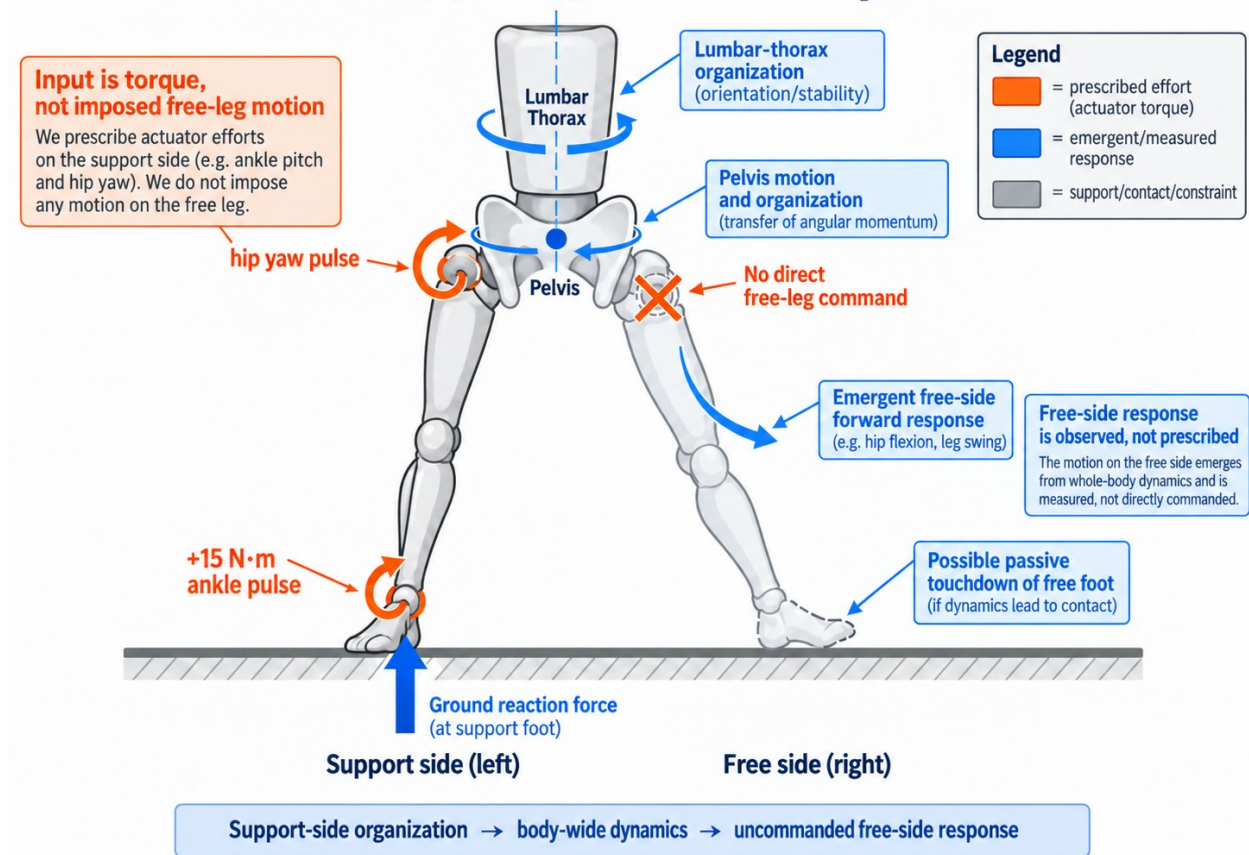


Figure 24. In the M2b logic, support-side torque is declared; the free side is not directly commanded.

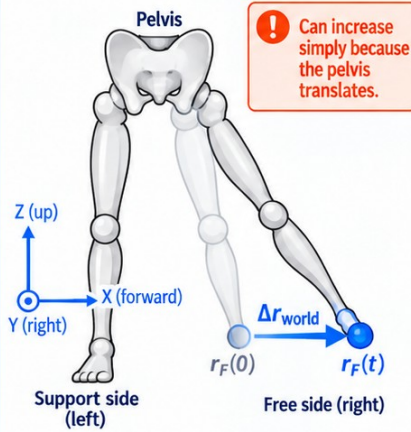
The core SupportInitiation package contains unitStep5, SmoothSupportPulse, HipDrive, AnkleDrive, LoadedSupportBody, M2bLoadedSupport and named baseline/ankle-only/hip-yaw/combined variants. The important scientific rule is to declare the effort source and keep the free-side response uncommanded when that response is the quantity under investigation.

25. “The free leg moved” is not a sufficient result

Three different free-leg measurements

1. World forward displacement

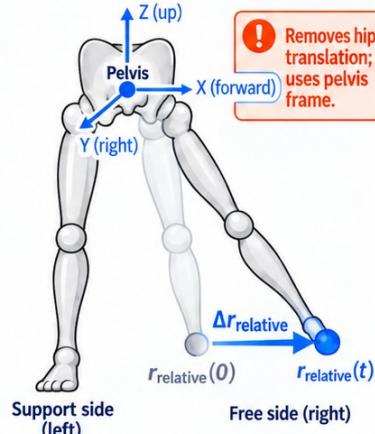
Ankle marker moves forward in room coordinates (e.g. global laboratory frame).



$$\Delta r_{\text{world}} = r_F(t) - r_F(0)$$

2. Pelvis-relative forward reach

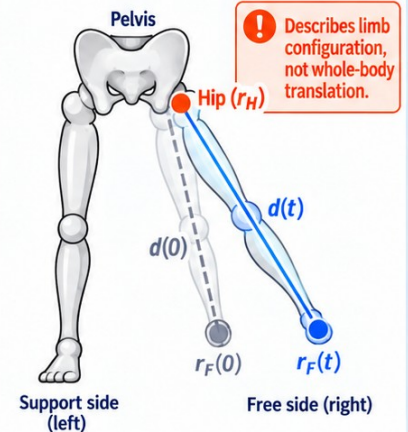
Ankle marker measure relative to the pelvis (current pelvis frame).



$$\Delta r_{\text{relative}} = r_{\text{relative}}(t) - r_{\text{relative}}(0)$$

3. Hip-to-ankle extension

Change in distance between hip and ankle markers (lengthening of the hip–ankle segment).



$$\Delta d = \|r_F - r_H\| - \|r_F(0) - r_H(0)\|$$

! These three quantities are not interchangeable.

1 World motion ≠ relative opening

Ankle can move forward in the world even if the hip–ankle distance does not change.

2 Relative opening ≠ extension alone

Pelvis-relative forward reach depends on both hip rotation and extension.

3 Always name the marker and frame

Specify which marker (e.g. ankle, hip) and which frame (world or pelvis) is used.

Figure 25. Three free-leg measurements answer different questions: world displacement, pelvis-relative reach and hip-to-foot distance change.

FOR DANCERS / GENERAL READERS

A foot can move forward in the room because the pelvis moved. That is not the same thing as the leg opening relative to the pelvis. It is also not the same thing as the hip-to-foot distance increasing. The guide therefore reports the frame and marker for every free-leg statement.

26. Contact validity and the later six-zone research layer

Contact validity: two evidence layers

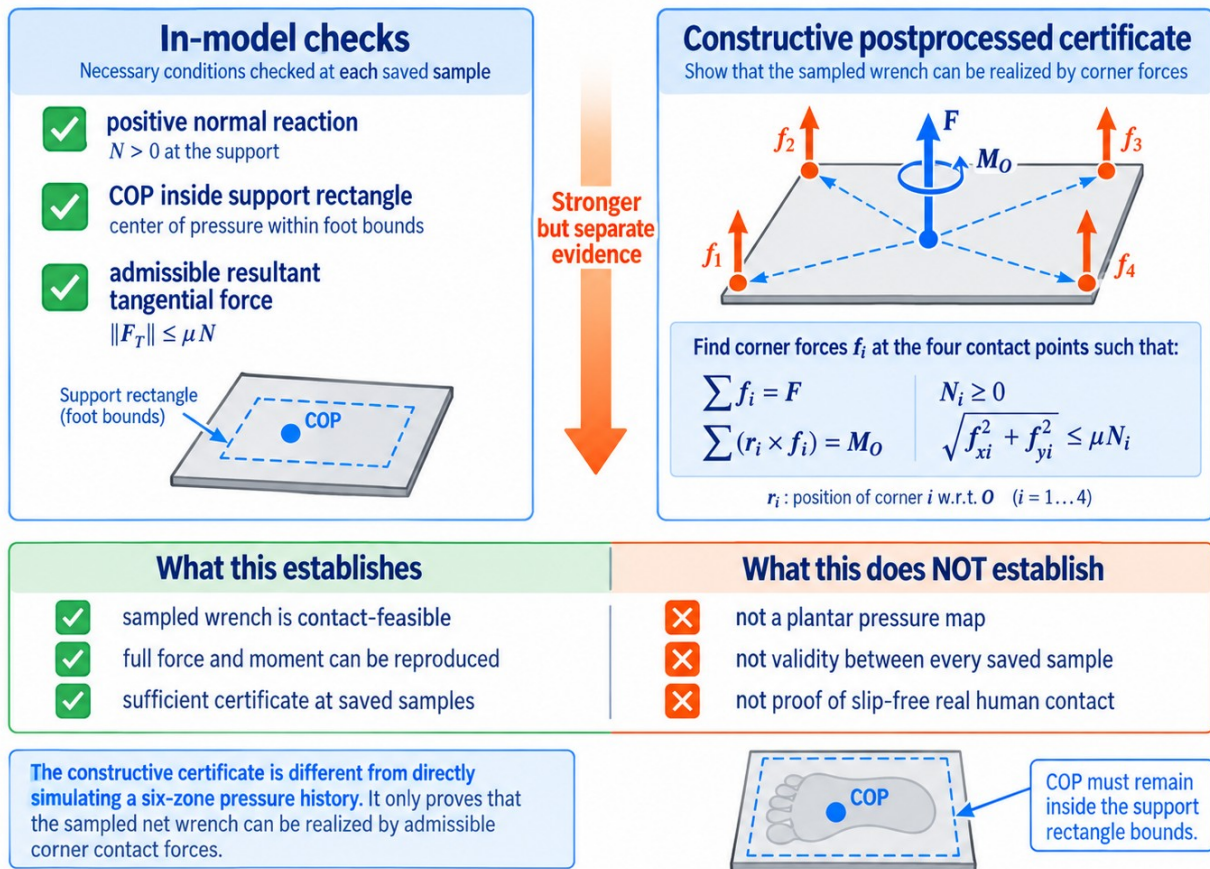


Figure 26. Fixed-support M2b contact admissibility and genuine load-bearing plantar contact are distinct evidence layers.

The historical fixed-support M2b remains useful for its declared question, but it cannot directly report heel/midfoot/forefoot load because an ideal support carries the standing reaction. The later contact-development branch uses actual six-zone support on both feet to study regional loading, passive touchdown and transfer. Those study runs are research-layer evidence rather than silently replacing the frozen core fixture.

NOW MEASURABLE	ADVERSE RESULT RETAINED	STILL OPEN
heel load, broad support, stance-axis progression, passive touchdown and new-foot support share	touchdown can occur while support organization later degrades; touchdown alone is not success	a clean fully loaded natural weight transfer with all acceptance gates simultaneously satisfied

WHAT THIS DOES NOT PROVE

The later contact-study branch is not human validation and not all of its development fixtures are promoted into the core v0.10 package.

27. Why lumbar architecture matters in practice

A controlled six-case comparison crossed locked, yaw-only and full 3-D lumbar architectures with right/left stance under the same body/contact/task conditions. Right-stance final free-side forward response was approximately 1.83 mm locked, 10.86 mm yaw-only and 10.13 mm full 3-D. The left-side cases mirrored signs closely.

FOR DANCERS / GENERAL READERS

For this tested rotational task, treating pelvis and thorax as one rigid block changed the result substantially. Allowing only axial lumbar yaw captured much of the full 3-D response. That is a model-reduction result for this task - not a claim that a dancer "should" move the spine in one specific way.

WHAT THIS DOES NOT PROVE

Architecture sensitivity is computational evidence. It does not assess an individual dancer's spinal mobility, health, technique quality or fitness.

PART VI - VERIFICATION, REPRODUCTION AND SAFE EXTENSION

A model that cannot be reproduced cannot support a strong article. The v0.10 release therefore separates runtime tests, build fixtures, analytical or known-answer contracts and publication-source hashing.

28. What 50/50 PASS means

v0.10 verification - what 50/50 PASS actually means

Verification protects a declared software/mechanics contract. It is not a realism score.



PASS = the declared numerical contract was reproduced in the verified environment.

Figure 27. The release register contains 45 runtime simulations and 5 build-only fixtures.

A PASS is local to a declared fixture. It means the expected numerical/mechanical property was reproduced within the test contract. It is not a global badge of realism.

Test family	What it protects
Geometry / axes	Neutral stance, hip, knee, ankle, pelvic ring and lumbar-axis direction checks.
Contact / wrench	Compression/separation, COP, reference-point/plane-height shifts, yaw decomposition, friction/spin and regional load checks.
Mass / data	Scaling smoke test, source tables, COM offset, affine mass properties, axial mass partition.
Dynamics / energy	Physical pendulum, passive dissipation, variable-impedance energy balance, spatial wrench power.
Research bridges	S45/S46 replay, standing-leg rotational transmission, matched impedance task and named support fixtures.
Build-only	Assembly translation/native executable creation without claiming a dynamic result.

29. The three v0.10-specific contracts

Test	Purpose
AuditedPlantarKnownLoadTest	Checks a known regional load split plus contact-energy identity.
AuditedPlantarHeelUnloadTest	Deliberately removes heel load; the observer must report the loss instead of inferring support from COP.
LoadedSupportVariableImpedanceEnergyTest	Checks the gravity-loaded aggregate ledger while stiffness changes and protects the modulation-energy sign.

WHAT THIS DOES NOT PROVE

A component can exist in the source and still be unused in a particular experiment. The load path of the fixture, not the class inventory alone, determines what was actually tested.

30. Reproduce the release

Reproduce the v0.10 release from a clean computer

The important result is not “it runs on my computer”; it is an auditable chain.



A future change should create a new version - never silently overwrite this source.

Figure 28. Clean reproduction workflow for the publication candidate.

Verified environment: OpenModelica 1.27.1 (64-bit) with Modelica Standard Library 4.1.0. Other translators/versions may work, but publication reproduction should record the exact toolchain.

Minimal load sequence

```
loadModel(Modelica,{"4.1.0"},requireExactVersion=true);
loadFile("ModelicaHumanBodyParts_v0_10_0_RC1.mo");
getVersion(ModelicaHumanBodyParts);
getErrorString();
```

A clean reproduction should preserve source hash, solver/tool versions, run/build logs, native executable/init artifacts where relevant, numerical CSV output, and checker results. Build-only fixtures must not be counted as runtime simulations.

31. Minimal use examples

Example A - resolve a segment

```
parameter Records.AnthropometryProfile p(...);
parameter Records.SegmentProperties thighProps = Scaling.resolveSegment(p, Types.SegmentKey.Thigh);
Segments.Thigh thigh(thighProps);
```

Example B - attach audited foot contact

```
Contact.AuditedPlantarContact contact(...);
for i in 1:6 loop
  connect(foot.frame_plantar[i], contact.plantarFrames[i]);
end for;
```

After simulation, inspect zoneNormalForce, heelLoad, midfootLoad, forefootLoad, heelShare, broadSupport, groundReactionForce, COP, loadedTangentialTravel and identityResidual.

Example C - variable impedance

```
Transmission.SmoothImpedanceSchedule schedule(...);
Transmission.VariableImpedance1D imp(kBase=..., cBase=..., qNeutral=...);
imp.stiffnessScale = schedule.scale;
imp.stiffnessScaleRate = schedule.scaleRate;
```

32. How to extend the library without breaking traceability

- Create a new version or branch; never silently overwrite a frozen publication source.
- For a new segment: define frame convention, mass/COM/inertia source and a known geometry/dynamics test before using it in a large assembly.
- For a new joint: define axis order and signs, then add an independent axis-geometry fixture.
- For new contact physics: include a known force/moment example, separation/recontact behavior and energy/passivity checks.
- For a new observer: verify it first against an analytical signal.
- For time-varying stiffness: keep modulation energy explicit.
- For a research experiment: declare effort source, boundary conditions, represented mass, measured outputs, success/failure gates and adverse outcomes before tuning parameters.

APPENDIX A - COMPLETE v0.10 COMPONENT DIRECTORY

The public chapters above focus on the body mechanics that dancers and general readers need first. This directory lists the reusable software components so an engineering reader can find the exact class/function family in the v0.10 source.

Types

Class / function	Plain-language role
Types.Sex	Selects the published coefficient branch where the data source distinguishes sex.
Types.SegmentKey	Names the modeled body segment requested by scaling functions.
Types.ProvenanceClass	Labels how a parameter or record entered the model.
Types.ScalingLevel	Records how far a parameter is from a directly measured input.
Types.ROMNature	Describes whether a mobility range is passive, active, assumed or otherwise classified.

Records

Class / function	Plain-language role
Records.AnthropometryProfile	The starting description of the modeled person.
Records.BSIPCoefficients	Body-segment inertial parameter coefficients from a chosen source table.
Records.SegmentProperties	Resolved mass, length, COM location and scalar principal inertias for a long segment.
Records.JointMobilityProfile	Lower and upper angle ranges for one or more joint coordinates.

Records.ProvenanceInfo	Compact source metadata carried with project parameters.
Records.SegmentProperties3D	Mass, full COM vector and full inertia tensor for an arbitrary rigid segment.
Records.JointMobilityMetadata	Describes the source, side and interpretation of a joint range.
Records.PublishedBSIPRow	A source-table row retained in a traceable machine-readable form.

Data

Class / function	Plain-language role
Data.DeLeva1996	Source-traceable anthropometric coefficient tables used by lower-body scaling.
Data.RoaasAndersson1982	Mean passive hip, knee and ankle mobility values from the documented source.
Data.Bova2020LowerLimb	Open-source lower-limb BSIP rows reproduced for source-audit comparisons.

Scaling

Class / function	Plain-language role
Scaling.segmentLength	Selects the explicit subject length associated with a SegmentKey.
Scaling.resolveSegment	Combines subject data with published BSIP coefficients to create a SegmentProperties record.

Utilities

Class / function	Plain-language role
Utilities.physicalInertia	Returns a physically usable inertia tensor while checking the required symmetry/positivity conditions.
Utilities.affineMassProperties	Transforms COM and inertia data when a segment frame is changed by a rigid affine mapping.
Utilities.shiftInertia	Moves an inertia tensor between parallel reference points.

Base

Class / function	Plain-language role
Base.PartialLongSegment	Reusable base for thigh and shank with proximal/distal MultiBody frames.
Base.RigidSegment3D	Rigid segment with a full COM vector and full inertia tensor.

Segments

Class / function	Plain-language role
------------------	---------------------

Segments.Thigh	Rigid femoral segment built from PartialLongSegment.
Segments.Shank	Rigid tibia/fibula segment built from PartialLongSegment.
Segments.Foot	Rigid foot segment with ankle frame, landmarks and six contact attachment points.
Segments.Pelvis	Rigid lower-trunk/pelvis proxy with fixed left and right hip-center frames.

Joints

Class / function	Plain-language role
Joints.Hip	Three serial revolute coordinates with side-aware axes and optional passive spring-damper behavior.
Joints.Knee	Single revolute flexion-extension coordinate with ROM monitoring.
Joints.AnkleComplex	Serial dorsiflexion/plantarflexion and inversion/eversion coordinates with side-aware signs.

Sensors

Class / function	Plain-language role
Sensors.JointLimitMonitor	Pure observer that reports how close each coordinate is to its declared range.

Contact

Class / function	Plain-language role
Contact.ContactPoint	One compliant ground point with non-tensile normal force and regularized Coulomb friction.
Contact.GroundWrenchSensor	Combines distributed contact forces and free moments into one resultant force, moment and COP.
Contact.PlantarContact	Six ContactPoint elements plus wrench reconstruction and optional residual spin moment.
Contact.torsionalCapacity	Diagnostic whole-foot RMS-radius capacity retained for regression and interpretation.
Contact.residualSpinCapacity	Allocates spin moment only from the local Coulomb traction that remains after translational shear.
Contact.TorsionalContactLaw	Applies a regularized yaw moment using only residual local traction capacity.
Contact.AuditedPlantarContact	v0.10 wrapper that adds regional loads and contact-energy auditing to PlantarContact.

Assemblies

Class / function	Plain-language role
Assemblies.LowerBodyAssembly	Legacy bilateral pelvis-legs assembly without ground contact.

Assemblies.LowerBodyGroundedAssembly	Legacy lower-body assembly with left and right PlantarContact models attached.
--------------------------------------	--

AxialBody

Class / function	Plain-language role
AxialBody.PelvicRing	3-D pelvic body with left/right hip centers and a lumbosacral connection.
AxialBody.LumbarSpine3D	Flexion/extension, lateral bend and axial rotation between pelvic ring and thorax.
AxialBody.Thorax	3-D upper-trunk rigid body with lumbar, cervical and shoulder reference frames.
AxialBody.PelvisLumbarThorax	PelvicRing + LumbarSpine3D + Thorax packaged as one reusable subassembly.
AxialBody.LowerBodyAxialAssembly	PelvicRing, 3-DOF lumbar, thorax and both articulated legs in one assembly.
AxialBody.LowerBodyAxialGroundedAssembly	Adds bilateral PlantarContact to the axial-body lower-body assembly.

Transmission

Class / function	Plain-language role
Transmission.VariableImpedance1D	Spring-damper element whose stiffness and damping scale smoothly with time, with explicit modulation energy.
Transmission.SmoothImpedanceSchedule	Quintic preparation-to-release scaling law with a continuous derivative.
Transmission.InterfaceLoadMetrics	Observer for force/moment magnitudes, impulses, loading rates, work and a high-load/low-motion diagnostic.
Transmission.KinematicSmoothnessMetrics	Computes acceleration magnitude, jerk vector/magnitude and accumulated jerk exposure.
Transmission.Knee	1-DOF knee with VariableImpedance1D plus reaction wrench and power closure outputs.
Transmission.AnkleComplex	Two ankle coordinates with variable impedance, reaction wrench and power outputs.
Transmission.Hip	Three hip coordinates with variable impedance and complete interface-wrench/power outputs.
Transmission.LumbarSpine3D	Three lumbar coordinates with variable impedance, reaction wrench and energy/power diagnostics.
Transmission.LowerBodyAxialAssembly	Axial body plus both legs with variable-impedance joints, COM outputs and seven interface-wrench channels.
Transmission.QuinticYawMission	Prescribed smooth rotational mission profile used in controlled comparisons.
Transmission.LiteratureRotationalTransmissionCase	Named experiment fixture for literature-informed pelvis/lumbar/thorax rotational transmission.
Transmission.SpatialWrenchPower	Small observer that combines a 3-D force/moment wrench with translational and angular velocity.

SupportInitiation

Class / function	Plain-language role
SupportInitiation.unitStep5	Normalized 0-to-1 fifth-order smooth step used by support pulses and schedules.
SupportInitiation.SmoothSupportPulse	Rise-hold-fall command envelope for support-side actuation.
SupportInitiation.HipDrive	Transmission.Hip plus three explicit torque-source inputs.
SupportInitiation.AnkleDrive	Transmission.AnkleComplex plus two explicit torque-source inputs.
SupportInitiation.LoadedSupportBody	Instrumented pelvis-lumbar-thorax + bilateral legs with declared ankle/hip-yaw actuation and variable-impedance controls.
SupportInitiation.M2bLoadedSupport	Gravity-loaded diagnostic with a fixed support boundary and passive free leg.
SupportInitiation.M2b variants	Four named configurations built from M2bLoadedSupport.

Benchmarks

Class / function	Plain-language role
Benchmarks.PartialFixture	Common parameter/boundary setup used by benchmark examples.
Benchmarks.FreeSwing	Simple gravity-driven segment motion used to check rigid-body dynamics.
Benchmarks.GroundedRelease	Grounded configuration released under gravity/contact for a small reproducible dynamic check.

Examples

Class / function	Plain-language role
Examples.LowerBodyFreeChainDemo	Minimal example showing how to instantiate the legacy lower-body assembly without ground contact.
Examples.LowerBodyRoaasMeanDemo	Example assembly using the source-traceable Roaas-Andersson mean mobility profiles.

CanonicalReplay

Class / function	Plain-language role
CanonicalReplay	Kinematic replay utilities and S45/S46 reference diagnostics retained as historical engineering comparators.

APPENDIX B - PUBLIC READING CHECKLIST

Before interpreting any figure or result on lynkorr.com, use this checklist.

- Name the exact model or experiment - do not merge M2a, M2b, contact development and human evidence.
- Name the standing side, coordinate frame and sign convention.
- Identify what is prescribed and what is allowed to emerge.
- Report the represented mass, not only the profile mass.
- For the foot, report absolute regional loads as well as percentages.
- For “free-leg opening,” name the marker and reference frame.
- For a transfer claim, require meaningful total vertical support in addition to a load-share ratio.
- For energy claims with variable stiffness, include the modulation-energy term.
- Retain adverse, null and infeasible cases.
- Do not turn software verification into physiological validation.

APPENDIX C - CURRENT LIMITS

These limits are part of the model description, not disclaimers added after the results.

- The body is a reduced rigid-segment model, not a deformable anatomical digital twin.
- Individual muscles, tendons, neural control, cartilage stress, ligament strain and injury probability are not identified.
- Six plantar zones are numerical support points, not a measured plantar-pressure grid.
- Regularized friction permits finite creep and does not represent exact static stiction.
- The lumbar spine is reduced to three relative rotations, not a vertebra-by-vertebra model.
- ROM records are diagnostic/source data and do not impose physiological hard stops.
- Time-varying impedance is a generic mechanical surrogate; no breathing or muscle physiology is inferred from it.
- Represented mass can differ from whole-body mass when anatomy is omitted.
- OpenModelica verification does not establish cross-translator reproduction until another translator is actually run.
- No current software result replaces force-plate, pressure-platform, motion-capture or other human validation.

Glossary for dancers and students

Term	Plain-language meaning
COM	Center of mass: the weighted average location of a rigid body or represented assembly mass.
DOF	Degree of freedom: one independent motion coordinate.
ROM	Range of motion: a declared/measured angular envelope; here it is mainly monitored, not mechanically enforced.
GRF	Ground reaction force: resultant force from the modeled floor on the modeled foot.
COP	Center of pressure: equivalent point on a reporting plane for the resultant wrench; not a unique pressure map.
Reaction wrench	Three forces + three moments transmitted through a mechanical interface.
Joint power	Generalized torque multiplied by generalized angular speed.
Spatial power	$F \cdot v + M \cdot \omega$ at a common point/frame.
Impedance	Here: a spring-damper relation between relative joint motion and torque; variable impedance means coefficients can change with time.
Verification	Evidence the implemented equation/software behavior matches a declared contract.
Validation	Evidence the model matches the real system sufficiently for the intended use.
Passive free leg	A free-side chain with no direct trajectory/position command in the experiment under discussion.

References and release record

- ModelicaHumanBodyParts v0.10.0 RC1 publication candidate source, SHA-256 2fda3b797ef9304529776d0610a3a8608e233c92531924579cb2eafe6665de4b.
- OpenModelica 1.27.1 (64-bit) and Modelica Standard Library 4.1.0 - verified release environment.
- de Leva, P. (1996). Adjustments to Zatsiorsky-Seluyanov's segment inertia parameters. Journal of Biomechanics 29(9), 1223-1230.
- Roaas, A. & Andersson, G.B.J. (1982). Normal range of motion of the hip, knee and ankle joints in male subjects, 30-40 years of age. Acta Orthopaedica Scandinavica 53, 205-208.
- Bova, G., Massaro, M. & Petrone, N. (2020). A Three-Dimensional Parametric Biomechanical Rider Model for Multibody Applications. Applied Sciences 10(13), 4509.
- Modelica Standard Library 4.1.0, Mechanics.MultiBody.
- The uploaded Deep Teaching / Complete Student Course Book served as the design and pedagogical model for this public edition; its core reading method - physical object, experiment, measured quantity, result, limitation - is preserved here.

WHAT THIS DOES NOT PROVE

All anatomical illustrations are teaching schematics supplied in the project material. They are not medical images and should not be interpreted as anatomical validation of the Modelica model.